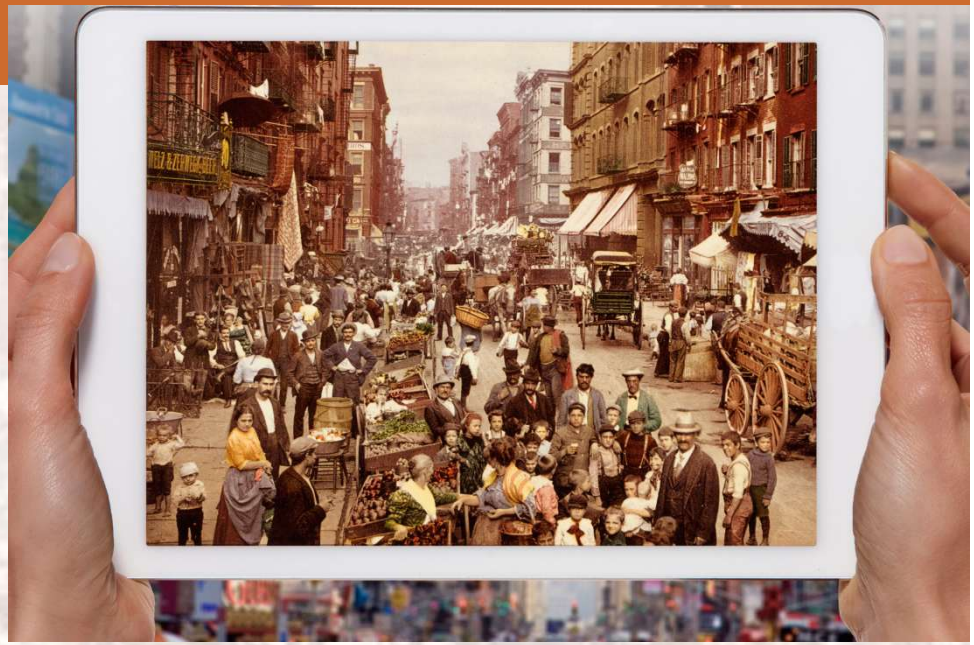
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CHAPTER 7

Consumers, Producers, and the Efficiency of Markets

National Central University

Look for the answers to these questions:

- What is consumer surplus? How is it related to the demand curve?
- What is producer surplus? How is it related to the supply curve?
- Do markets produce a desirable allocation of resources? Or could the market outcome be improved upon?

This chapter discusses a perfectly competitive market



Welfare Economics

- Allocation of resources refers to:
  - How much of each good is produced
  - Which producers produce it
  - Which consumers consume it
- Welfare economics
  - Studies how the allocation of resources affects economic well-being

Welfare economics looks at:
- Benefits that buyers and sellers receive from engaging in market transactions
- How society can make these benefits as large as possible

In any market, the equilibrium of supply and demand maximizes the total benefits received by all buyers and sellers combined



Resource Allocation Methods

Scarce resources might be allocated by

- Market price
- Command
- Majority rule
- Contest
- First-come, first-served
- Sharing equally
- Lottery
- Force

How does each method work?



Willingness to Pay (WTP) 願付價格

- A buyer's willingness to pay for a good
  - Maximum amount the buyer will pay for that good
  - How much the buyer values the good

name	WTP
Anthony	$250
Chad	175
Flea	300
John	125

Example:
4 buyers' WTP
for an iPod



WTP and the Demand Curve

Q: If price of iPod is $200, who will buy an iPod, and what is quantity demanded?

name	WTP
Anthony	$250
Chad	175
Flea	300
John	125

A: Anthony & Flea will buy an iPod, Chad & John will not.

Hence, $Q^d = 2$ when $P = \$200$.



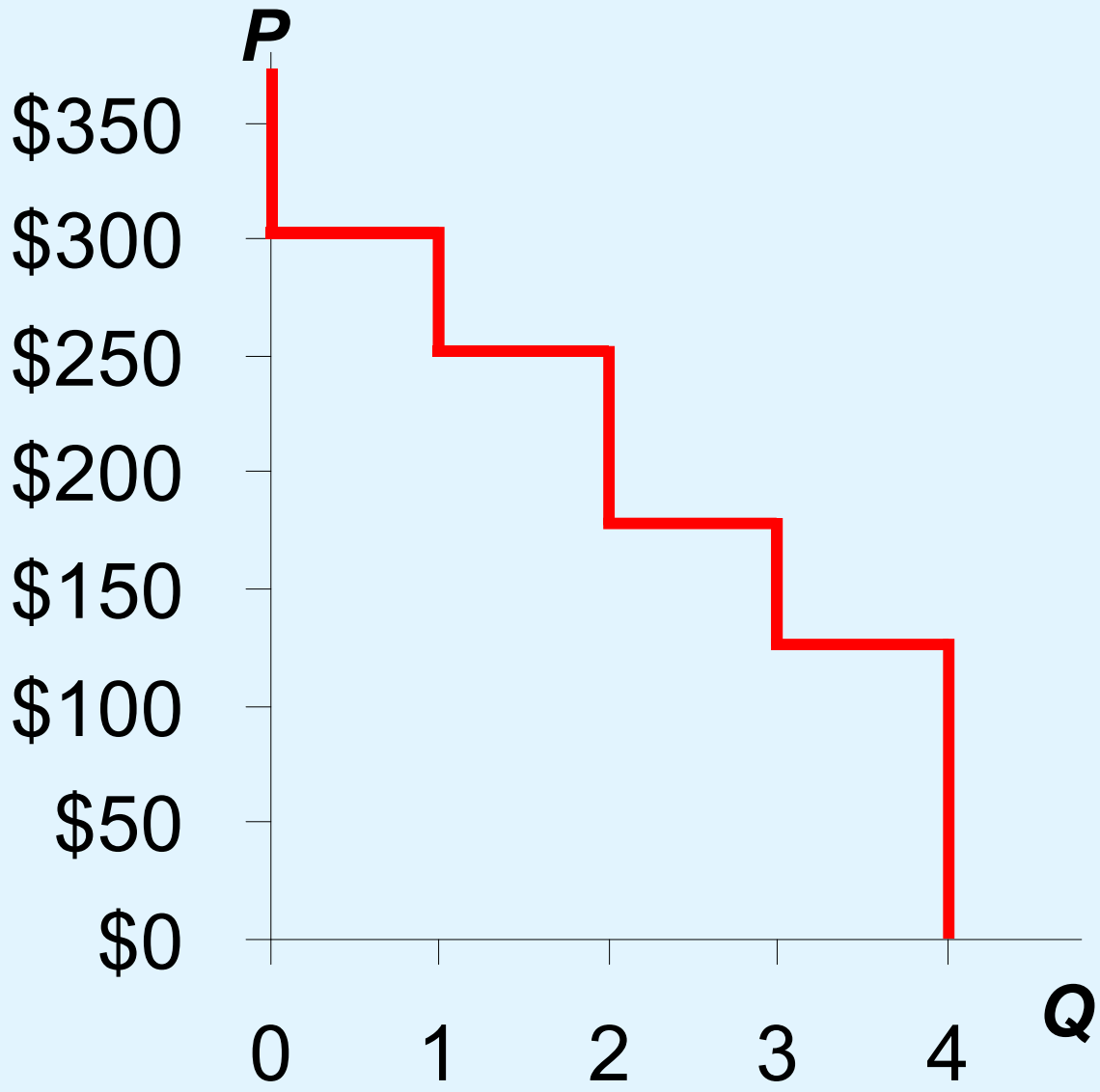
WTP and the Demand Curve

- Derive the demand schedule:

name	WTP
Anthony	$250
Chad	175
Flea	300
John	125

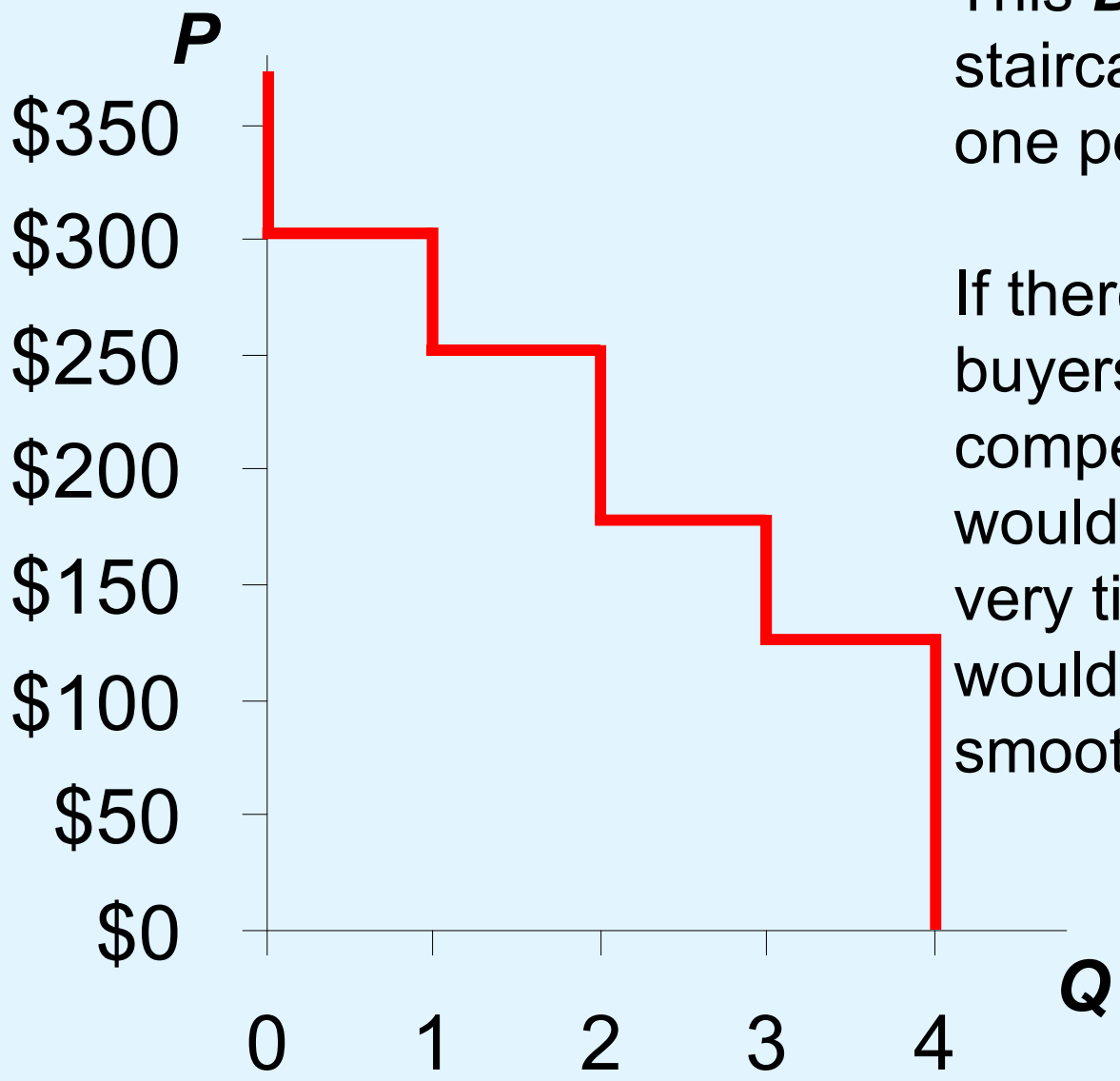
P (price of iPod)	who buys	Q^d
$301 & up	nobody	0
251 – 300	Flea	1
176 – 250	Anthony, Flea	2
126 – 175	Chad, Anthony, Flea	3
0 – 125	John, Chad, Anthony, Flea	4

WTP and the Demand Curve



P		Q^d
$301 & up		0
251 –	300	1
176 –	250	2
126 –	175	3
0 –	125	4

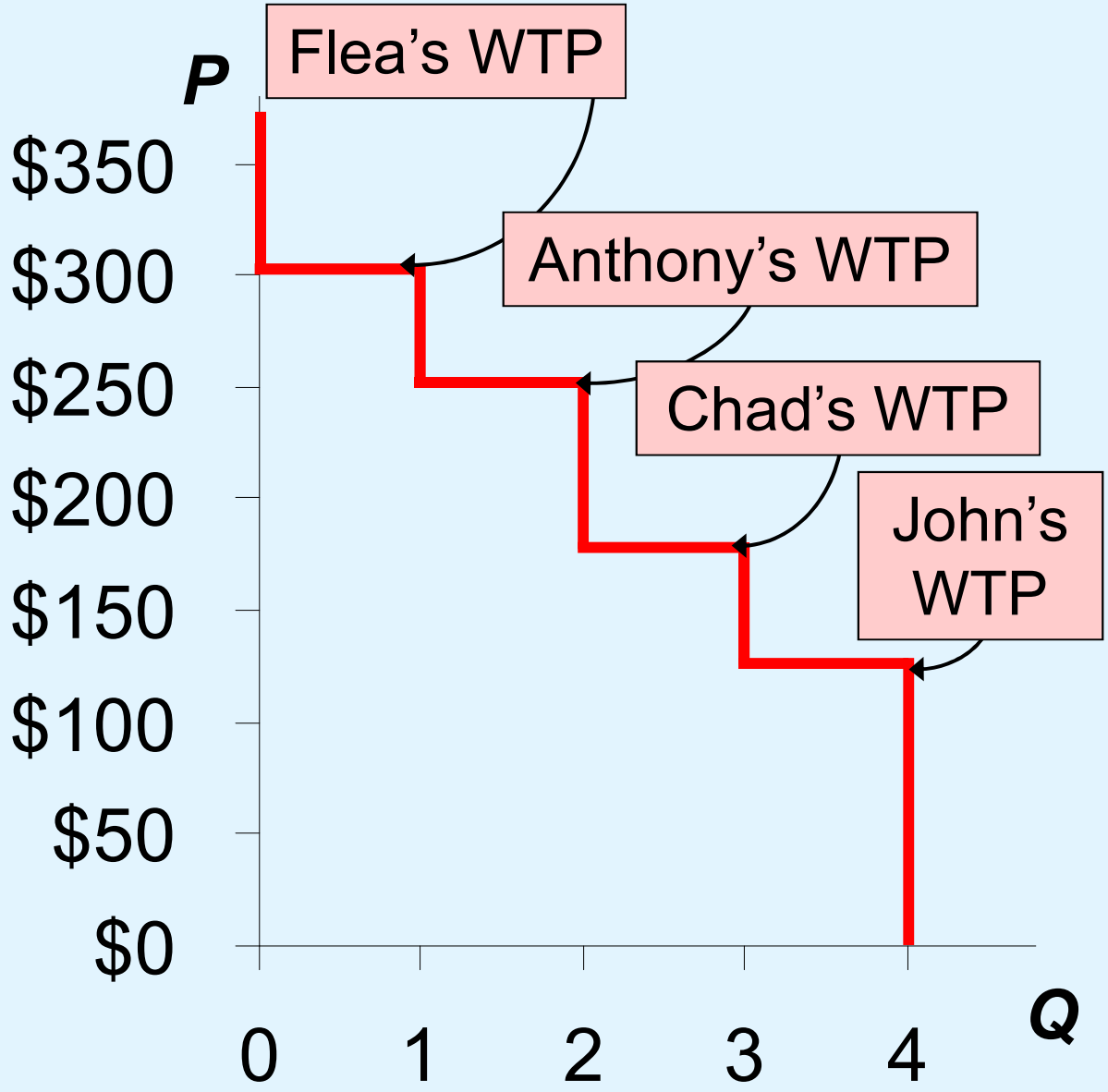
About the Staircase Shape…



This ***D*** curve looks like a staircase with 4 steps – one per buyer.

If there were a huge # of buyers, as in a competitive market, there would be a huge # of very tiny steps, and it would look more like a smooth curve.

WTP and the Demand Curve



At any **Q**, the height of the ***D*** curve is the WTP of the ***marginal buyer***, the buyer who would leave the market if ***P*** were any higher.

name	WTP
Anthony	$250
Chad	175
Flea	300
John	125



Consumer Surplus (CS) 消費者剩餘

- Consumer surplus CS = WTP – P
  - Amount a buyer is willing to pay minus the amount the buyer actually pays:

name	WTP
Anthony	$250
Chad	175
Flea	300
John	125

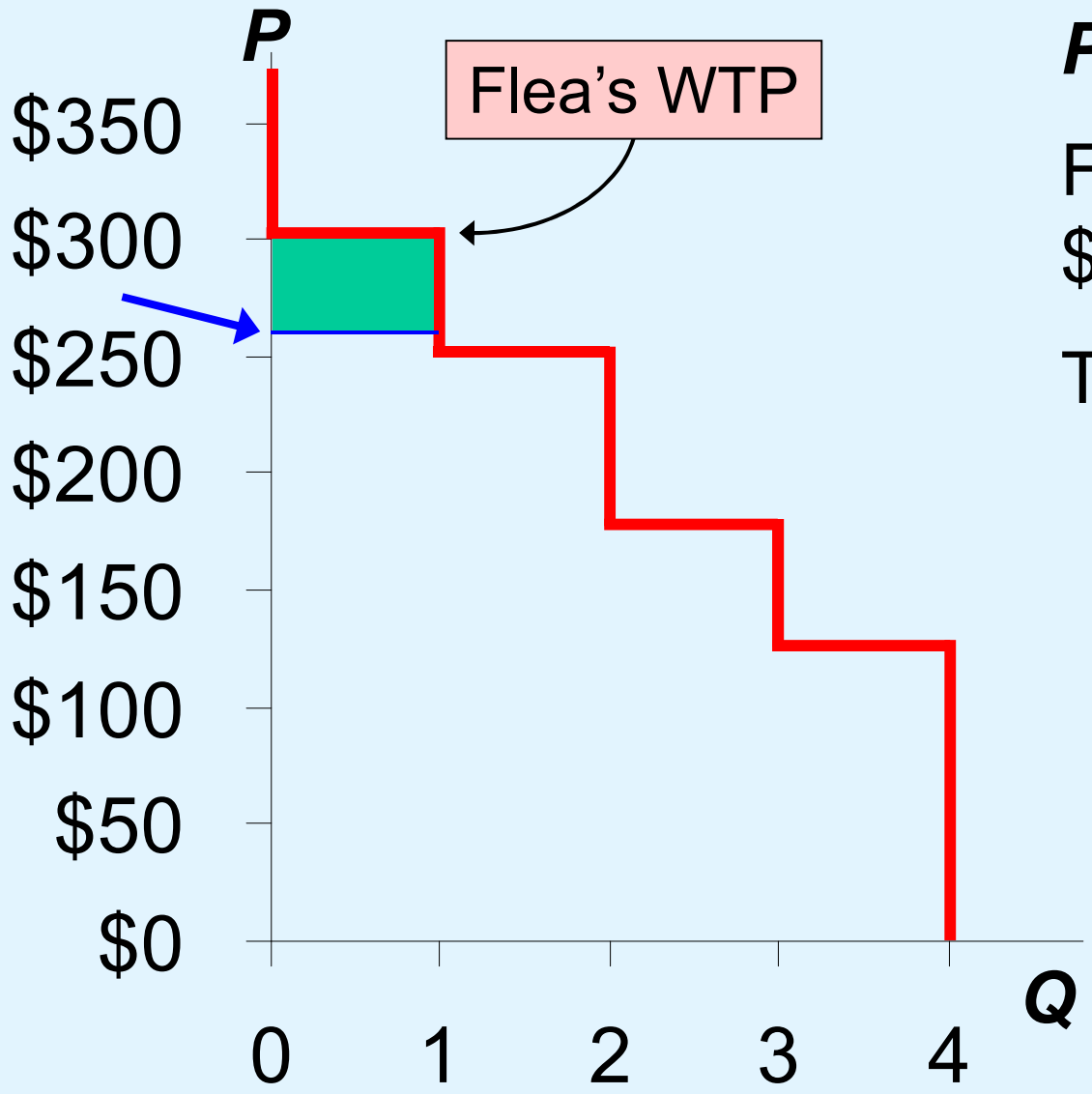
Suppose P = $260.

Flea's CS = $300 – 260 = $40.

The others get no CS because they do not buy an iPod at this price.

Total CS = $40.

CS and the Demand Curve

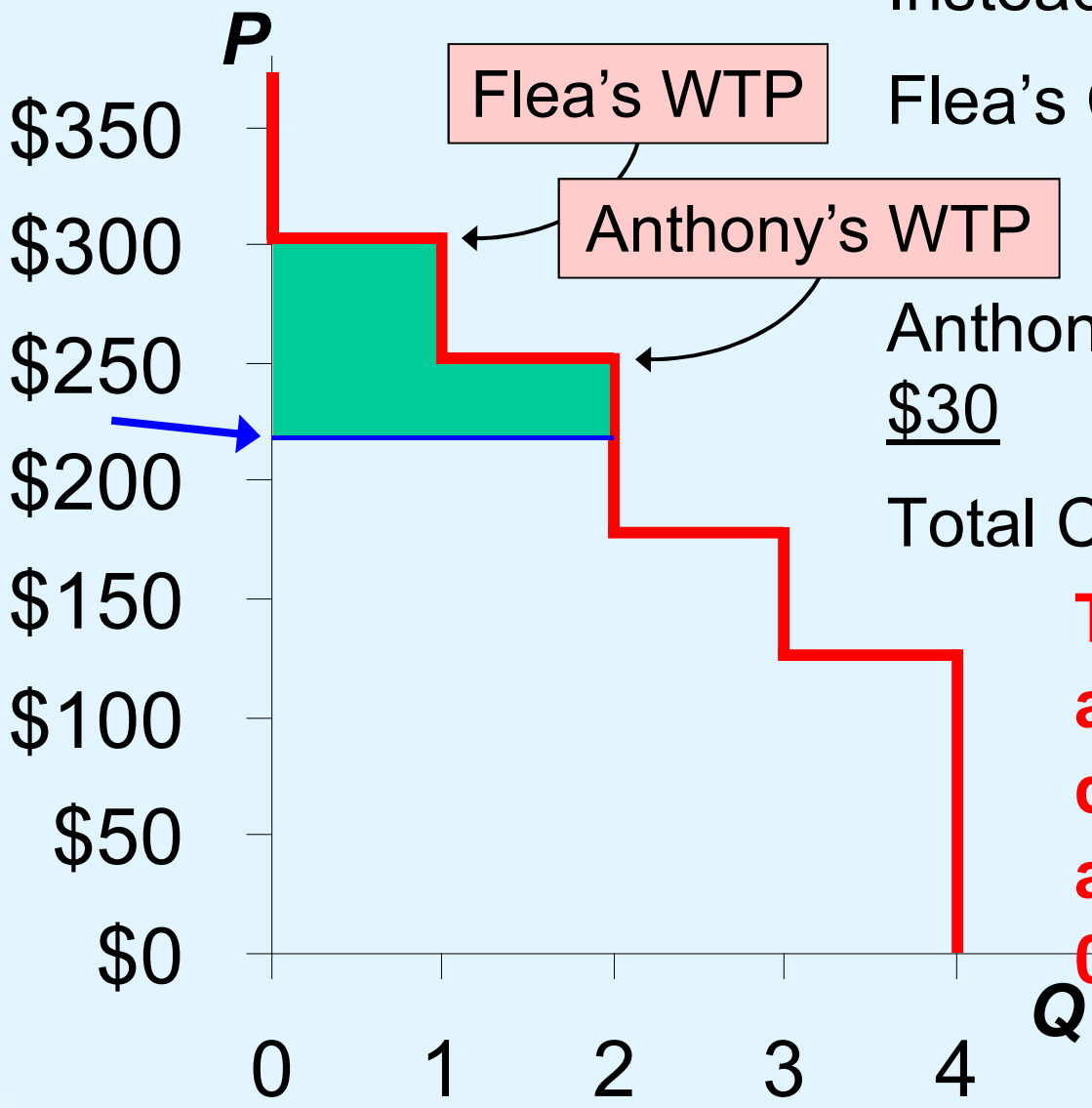


P = $260

Flea's CS =
$300 – 260 = $40

Total CS = $40

CS and the Demand Curve



Instead, suppose ***P*** = $220

Flea's CS = $300 – 220 = $80

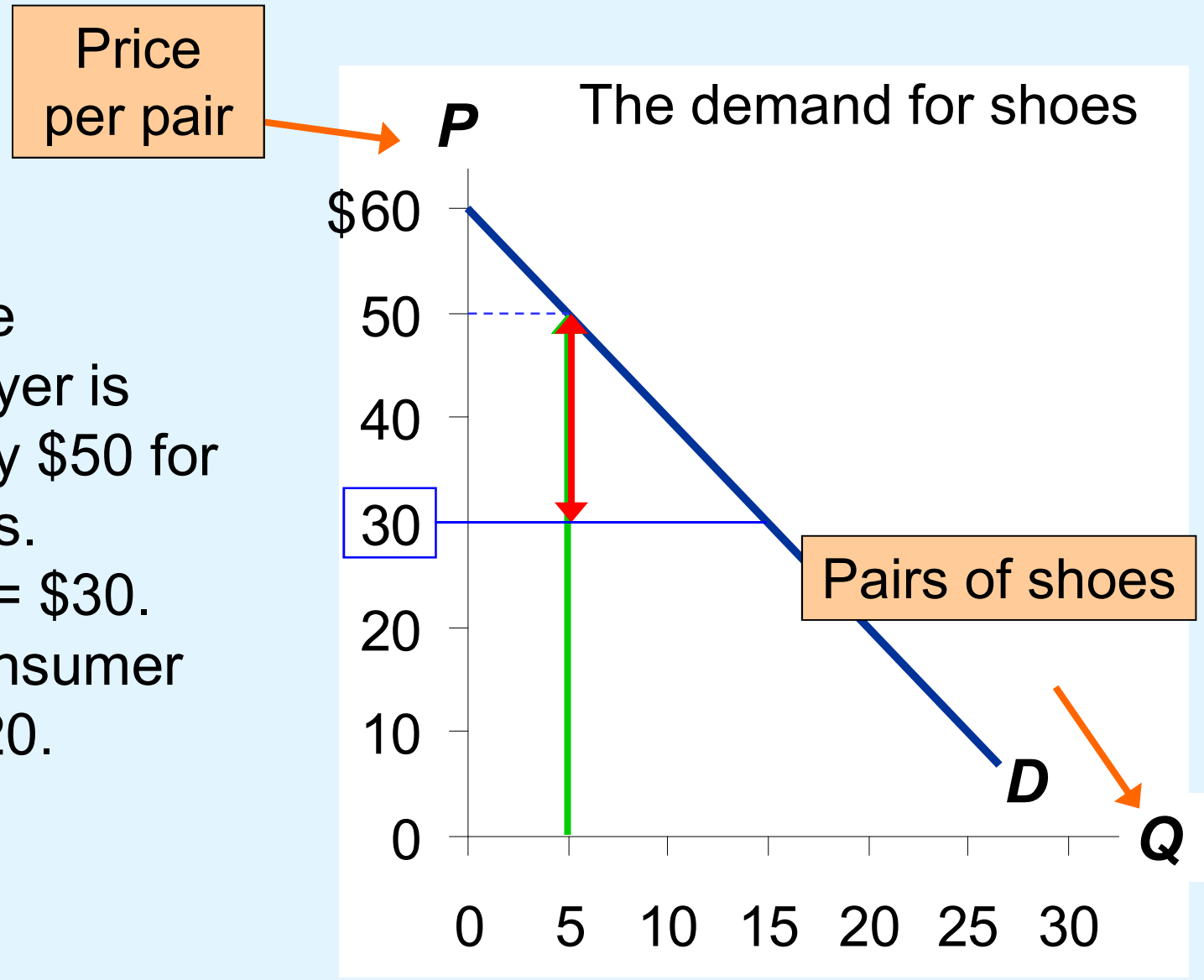
Anthony's CS = $250 – 220 = $30

Total CS = $110

Total CS equals the area under the demand curve above the price, from 0 to Q.

CS with Lots of Buyers & a Smooth D Curve

At ***Q*** = 5, the marginal buyer is willing to pay $50 for pair of shoes. Suppose ***P*** = $30. Then his consumer surplus = $20.

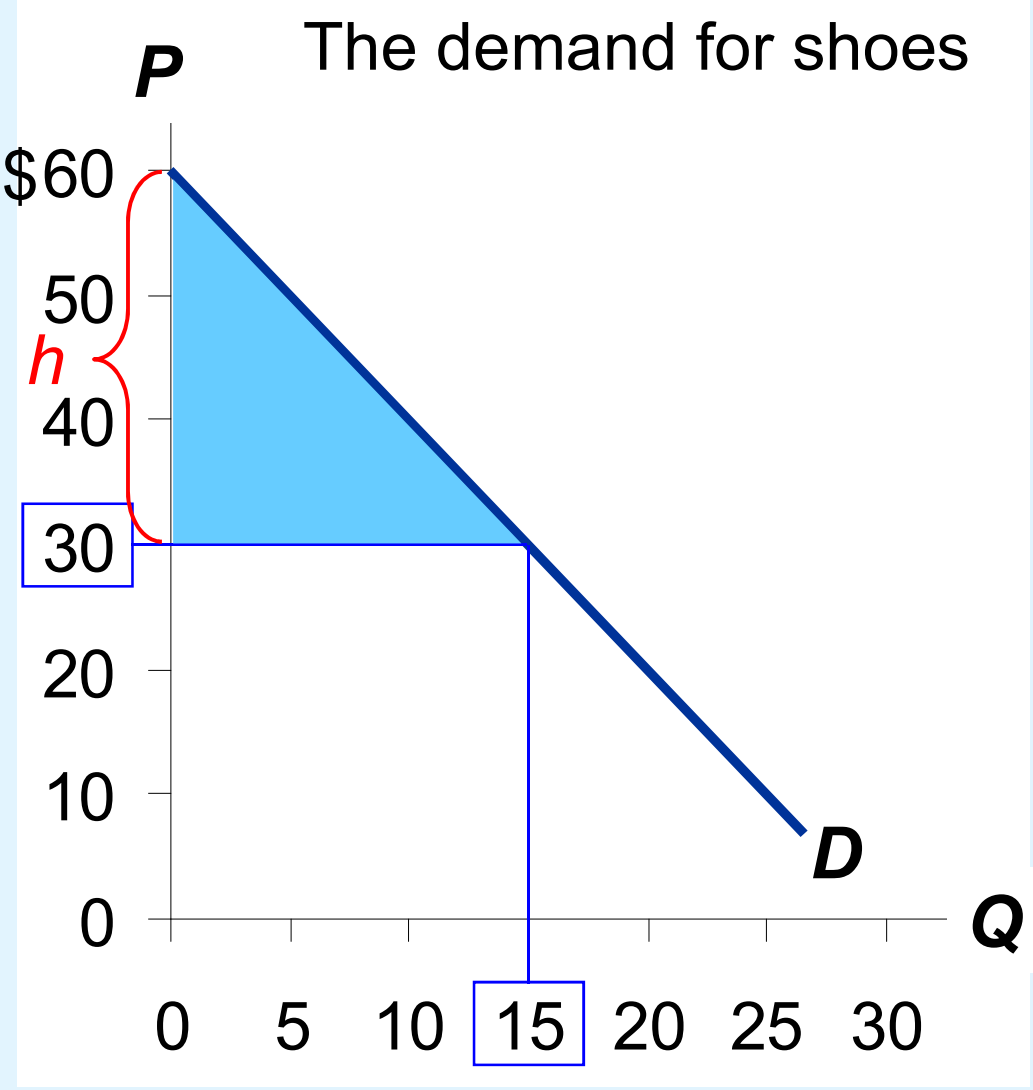


CS with Lots of Buyers & a Smooth D Curve

CS is the area between ***P*** and the ***D*** curve, from 0 to ***Q***.

Recall: area of a triangle equals ½ x base x height

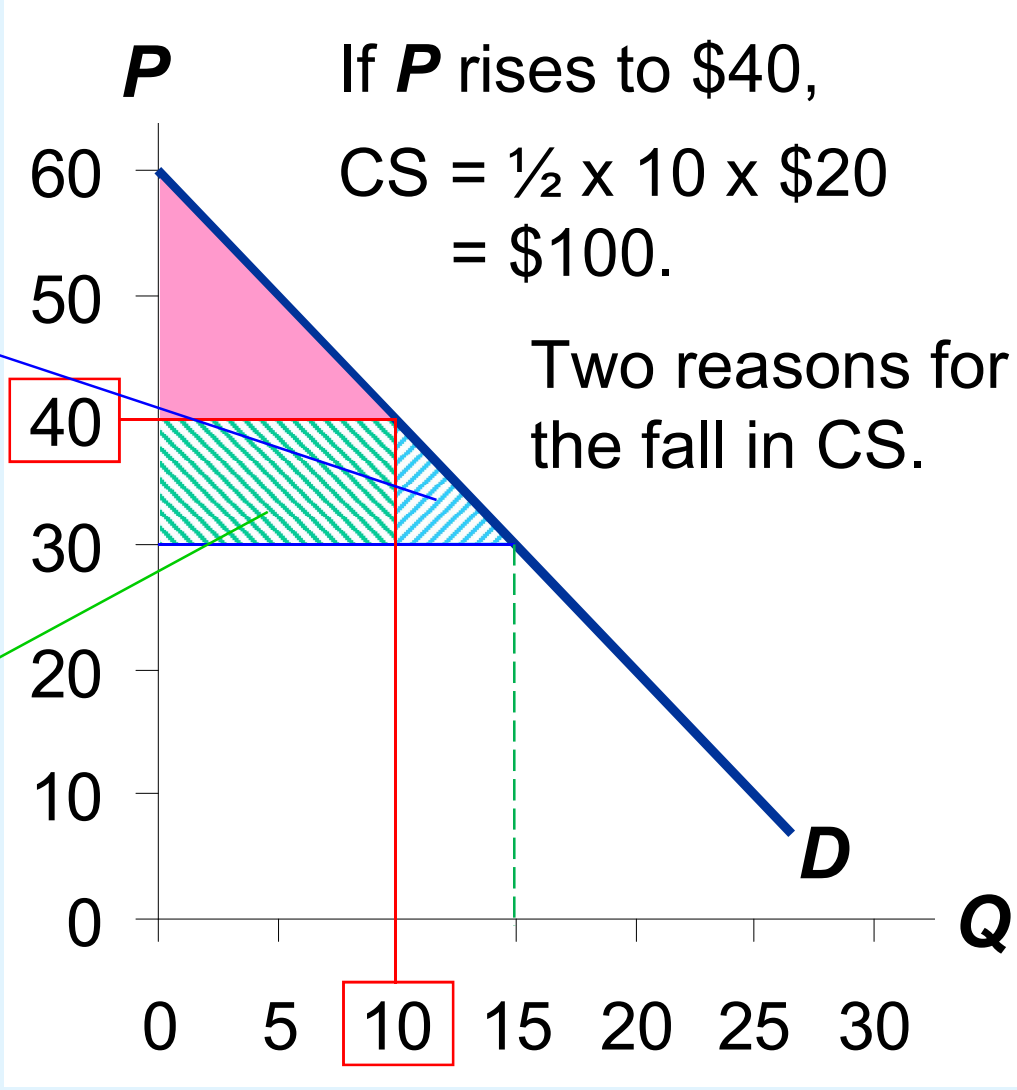
Height = $60 – 30 = $30.
So,
CS = ½ x 15 x $30
= $225.



How a Higher Price Reduces CS

1. Fall in CS due to buyers leaving market

2. Fall in CS due to remaining buyers paying higher ***P***



If ***P*** rises to $40,

CS = ½ x 10 x $20
= $100.

Two reasons for the fall in CS.

Active Learning 1

Consumer surplus

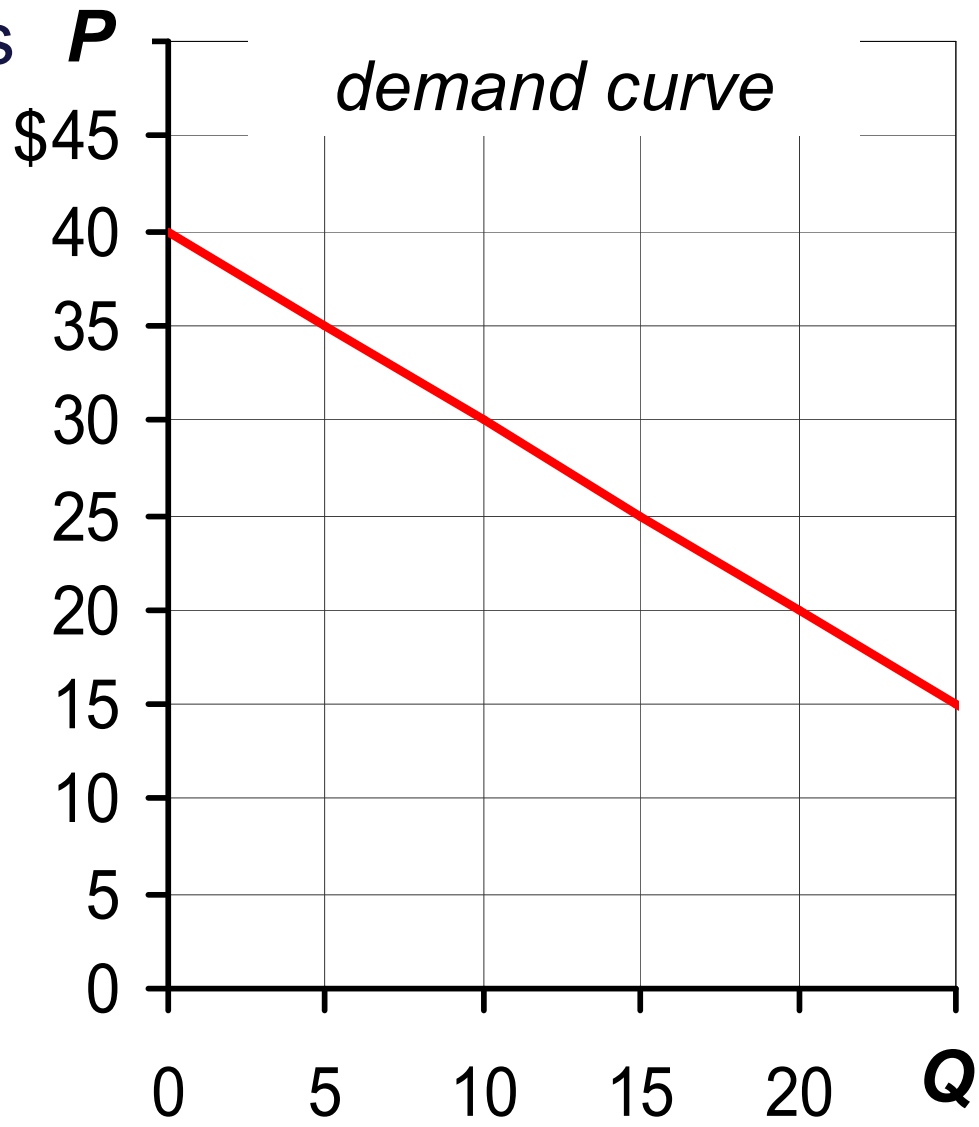
A. Find marginal buyer's WTP at Q = 10.

B. Find CS for P = $30

Suppose ***P*** falls to $20. How much will CS increase due to…

C. buyers entering the market

D. existing buyers paying lower price



Active Learning 1

Answers

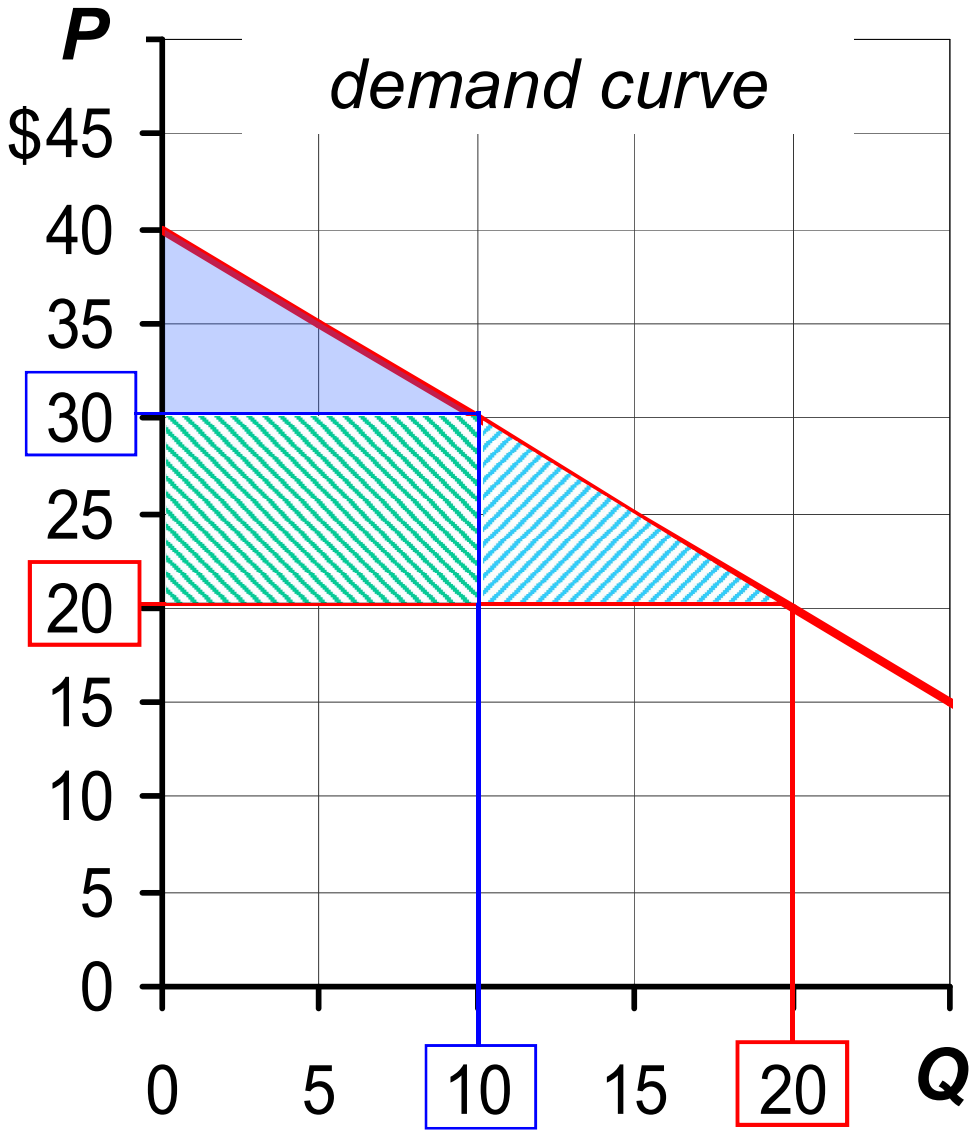
A. At Q = 10, marginal buyer's WTP is $30.

B. CS = ½ x 10 x $10 = $50

P falls to $20.

C. CS for the additional buyers = ½ x 10 x $10 = $50

D. Increase in CS on initial 10 units = 10 x $10 = $100



Question

Melissa buys an iPhone for $240 and gets consumer surplus of $160

a. What is her willingness to pay?
b. If she had bought the iPhone on sale for $180, what would her consumer surplus have been?
c. If the price of an iPhone were $500, what would her consumer surplus have been?



Producer Surplus 生產者剩餘

- Cost
  - Value of everything a seller must give up to produce a good
    - Measure of willingness to sell: produce and sell the good/service only if the price > cost

name	*cost*
Jack	$10
Janet	20
Chrissy	35

Example: Costs of 3 sellers in the lawn-cutting business.

The cost includes cost of all resources used to produce good, including value of the seller's time (opportunity cost).



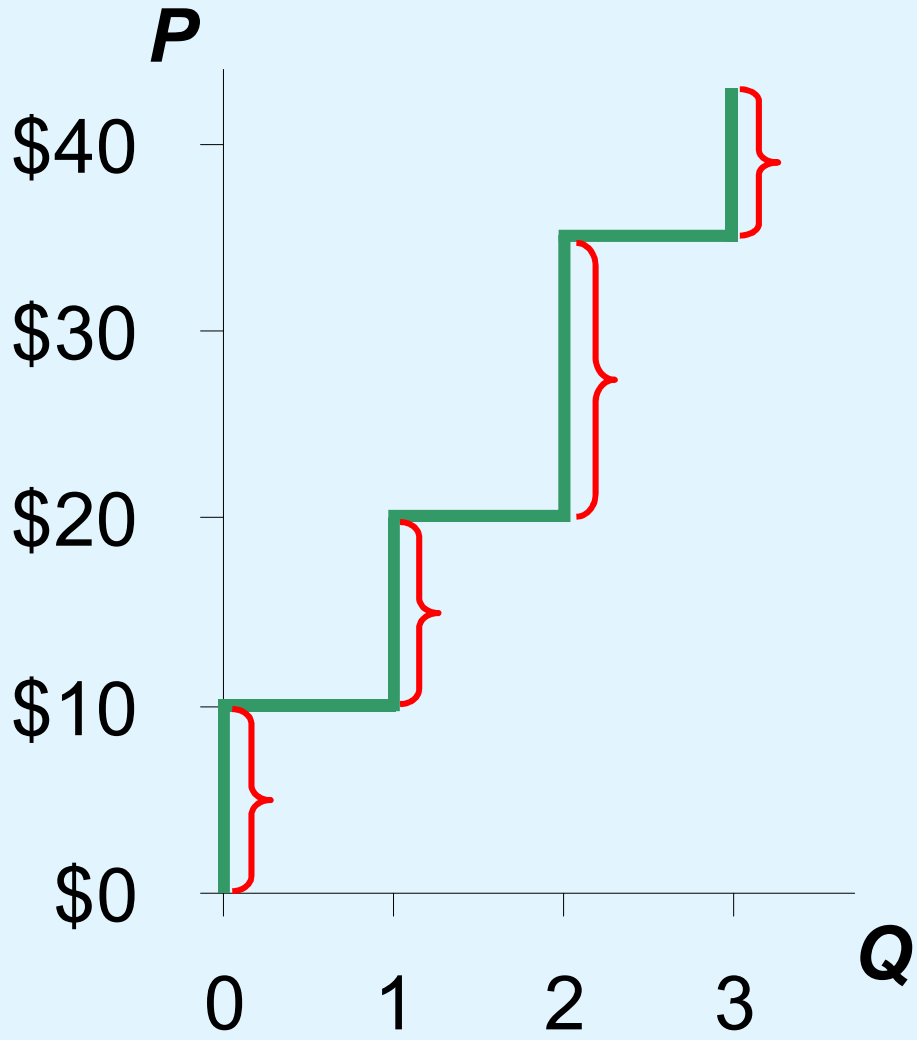
Producer Surplus

- Derive the supply schedule from the cost data:

name	*cost*
Jack	$10
Janet	20
Chrissy	35

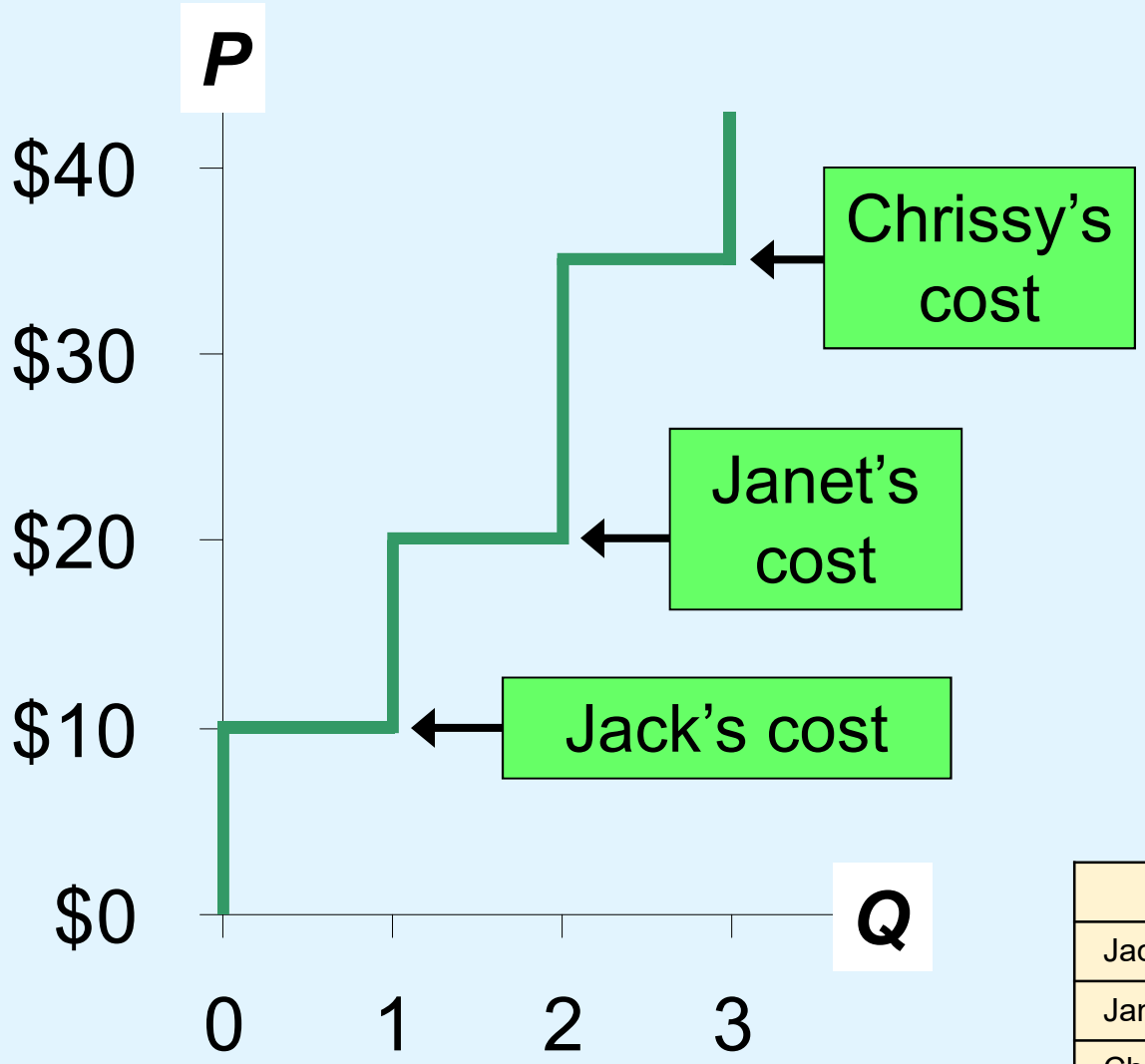
P	Q^s
$0 – 9	0
10 – 19	1
20 – 34	2
35 & up	3

Cost and the Supply Curve



P	Q^s
$0 – 9	0
10 – 19	1
20 – 34	2
35 & up	3

Cost and the Supply Curve



At each ***Q***, the height of the ***S*** curve is the cost of the ***marginal seller***, the seller who would leave the market if the price were any lower.

name	*cost*
Jack	$10
Janet	20
Chrissy	35

We can use the term "marginal cost" as short-hand for "cost of the marginal seller."

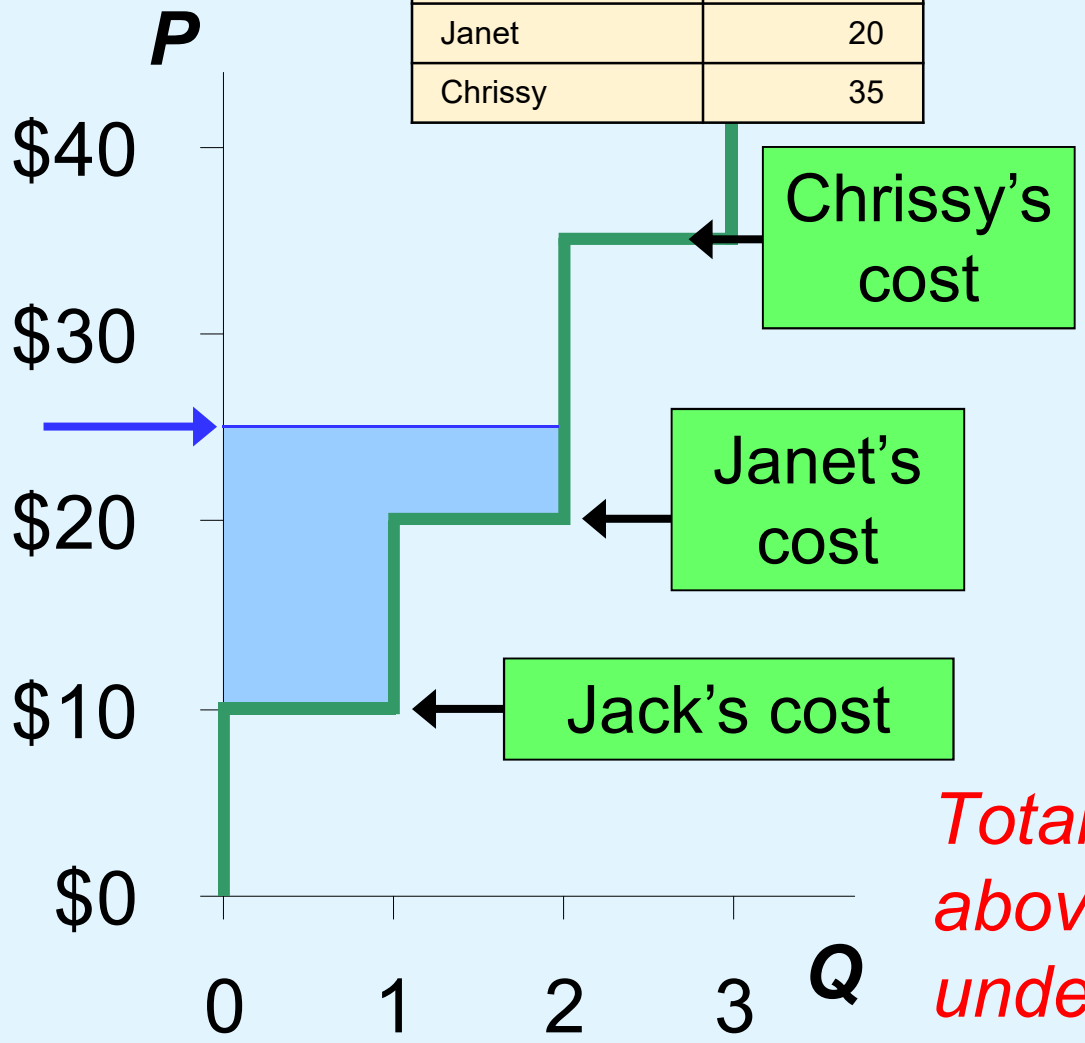


Producer Surplus

- Producer surplus, PS = P - cost
  - Amount a seller is paid for a good minus the seller's cost of providing it
  - Price received minus willingness to sell

Producer Surplus and the S Curve

name	*cost*
Jack	$10
Janet	20
Chrissy	35



PS = ***P*** – cost

Suppose ***P*** = $25.

Jack's PS = $15

Janet's PS = $5

Chrissy's PS = $0

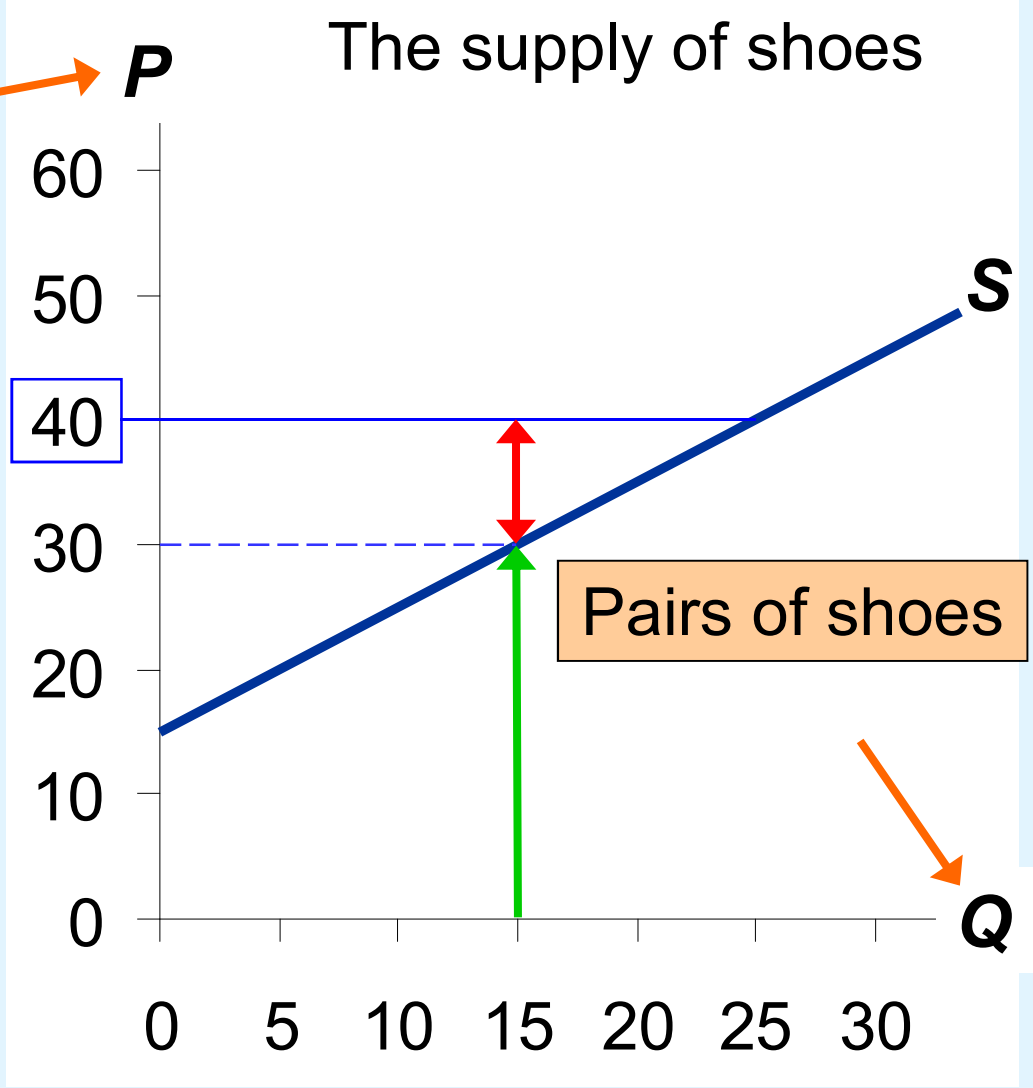
Total PS = $20

Total PS equals the area above the supply curve under the price, from 0 to Q.

PS with Lots of Sellers & a Smooth S Curve

Suppose ***P*** = $40.

At ***Q*** = 15, the marginal seller's cost is $30, and her producer surplus is $10.

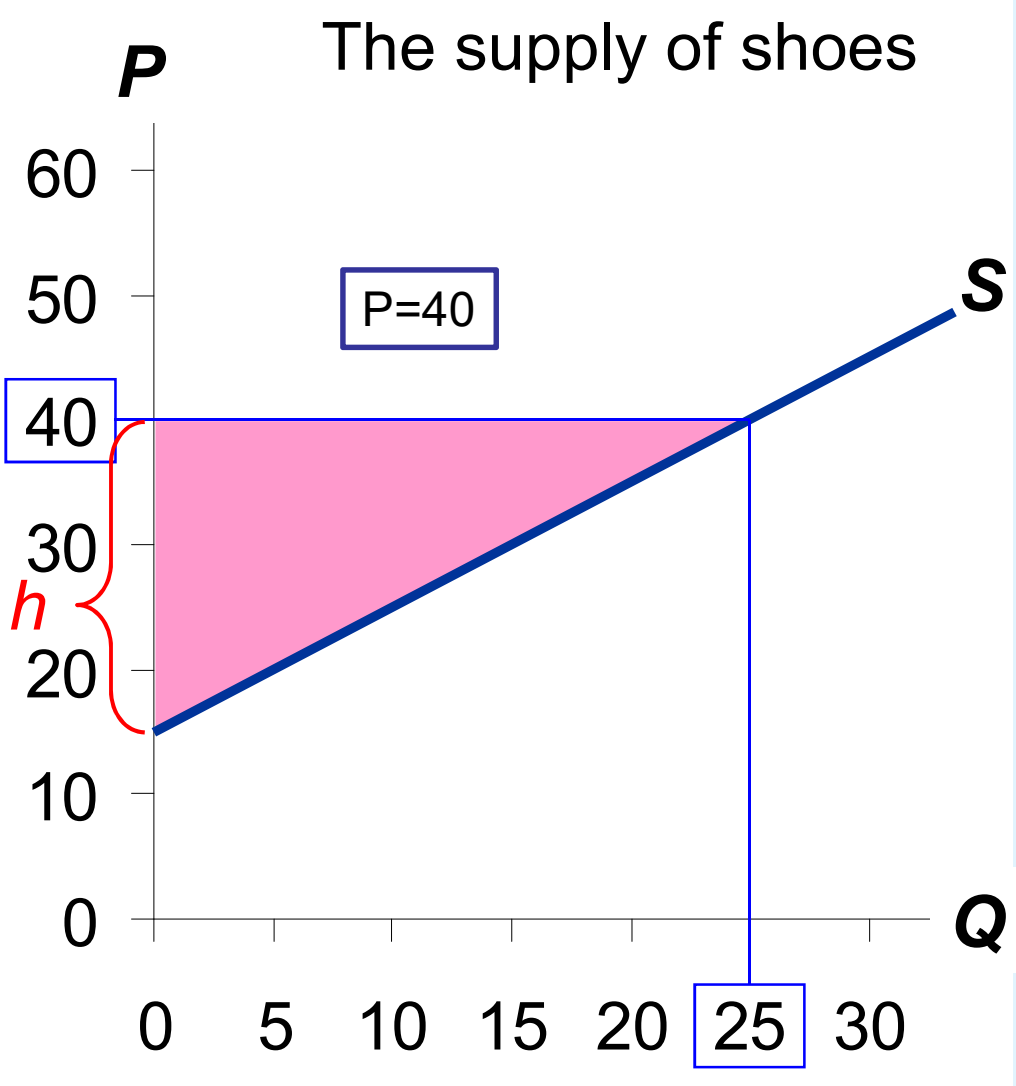


PS with Lots of Sellers & a Smooth S Curve

PS is the area between ***P*** and the ***S*** curve, from 0 to ***Q***.

The height of this triangle is
$40 – 15 = $25.

So,
PS = ½ x *b* x *h*
= ½ x 25 x $25
= $312.50

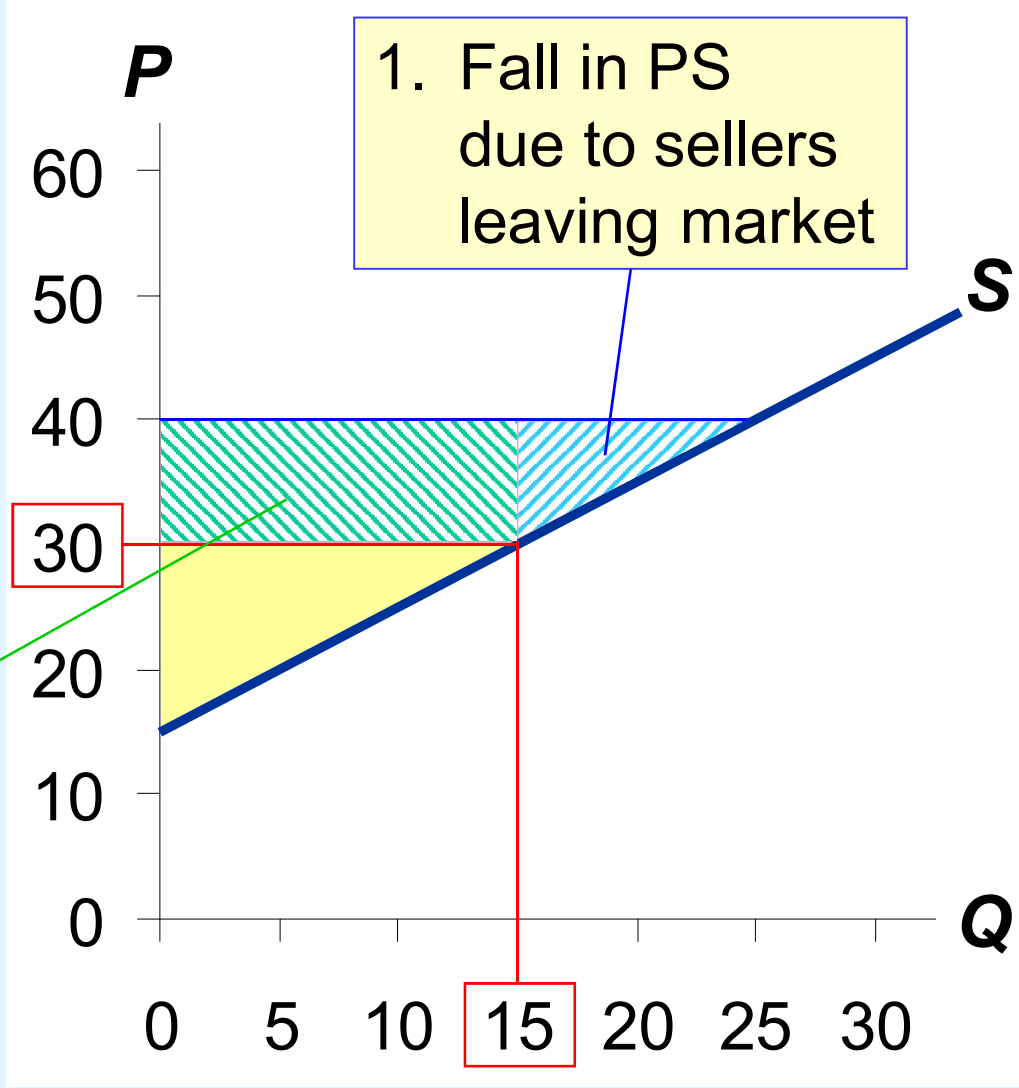


How a Lower Price Reduces PS

If ***P*** falls to \$30,

PS = ½ x 15 x \$15
= \$112.50

Two reasons for the fall in PS.



Active Learning 1

A. Find marginal seller's cost at Q = 10.

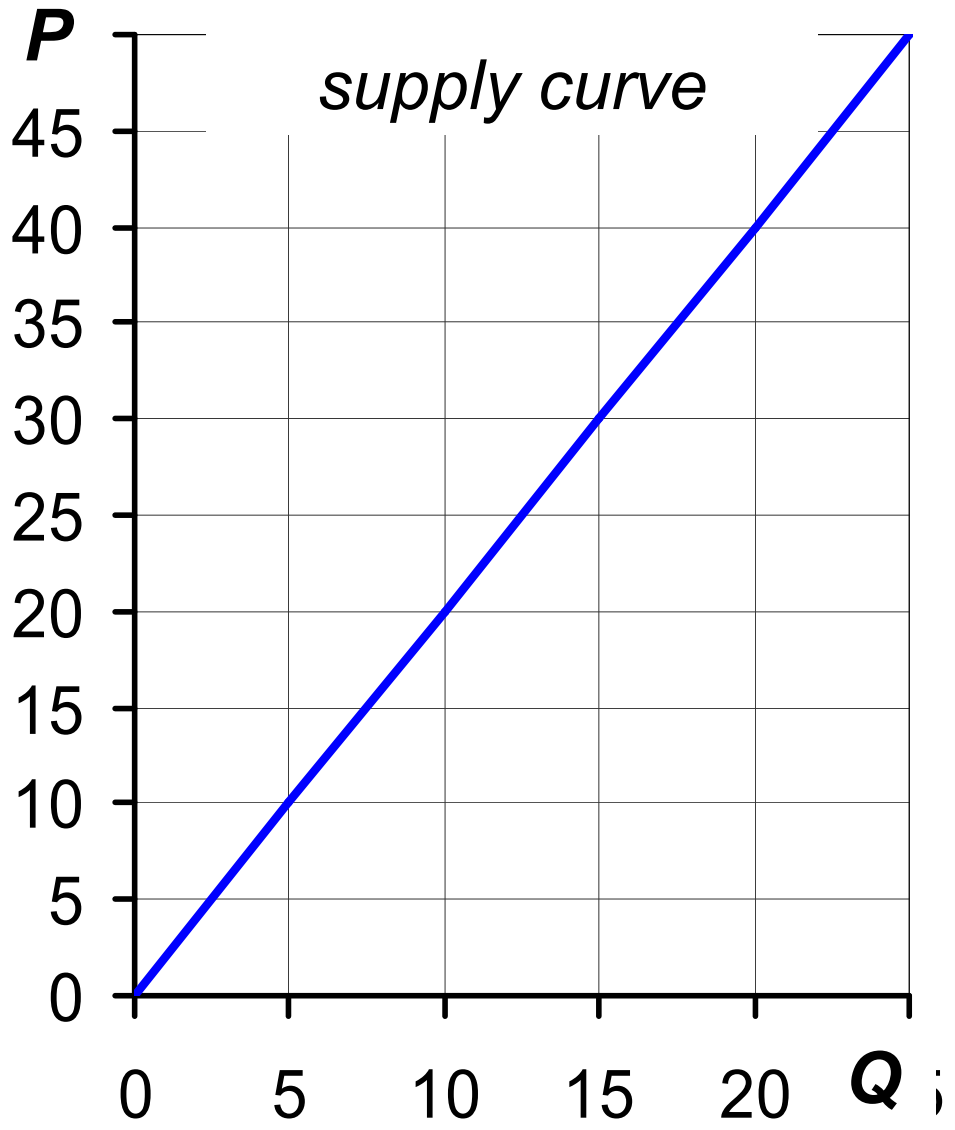
B. Find total PS for P = $20.

Suppose P rises to $30. Find the increase in PS due to:

C. selling 5 additional units

D. getting a higher price on the initial 10 units

Producer surplus



Active Learning 1

A. At Q = 10,
marginal cost = $20

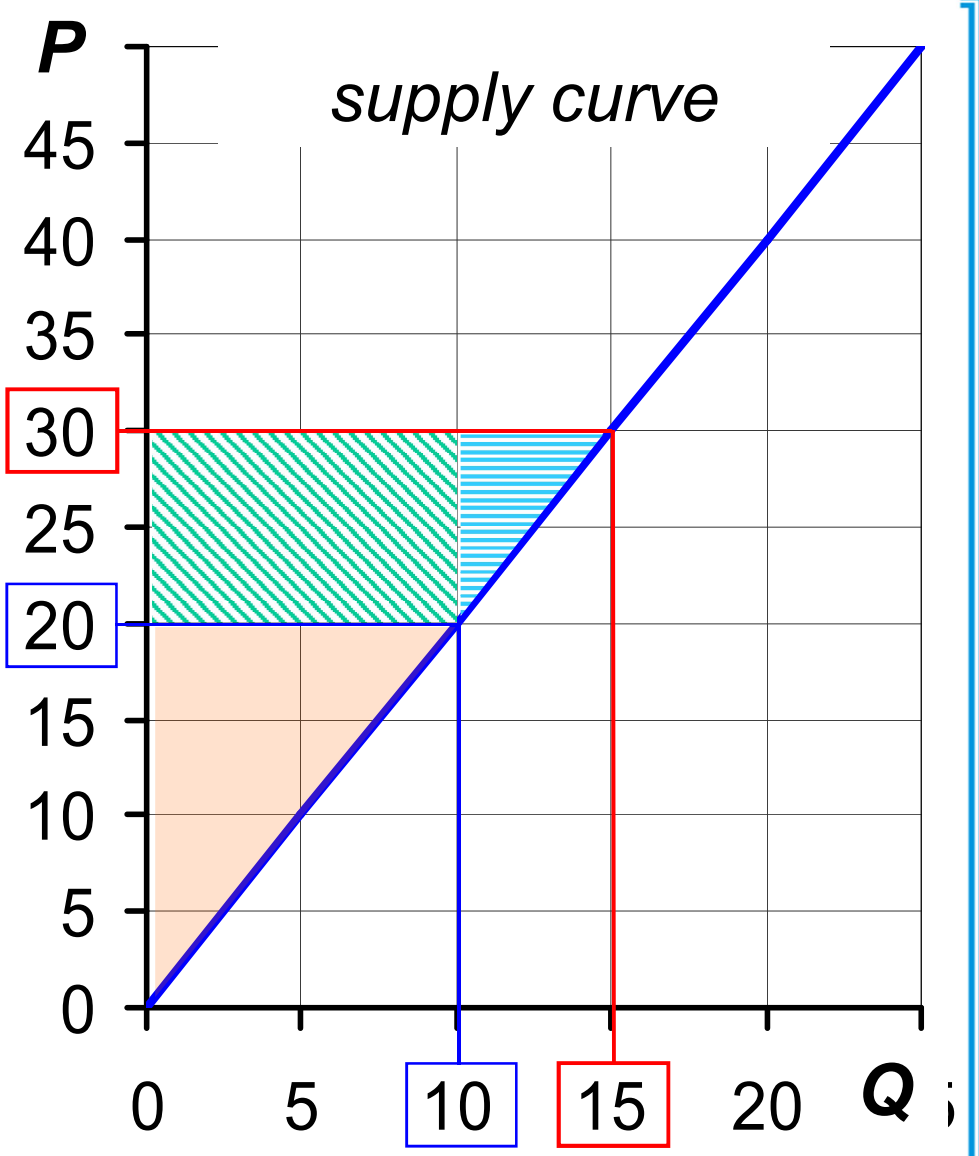
B. PS = ½ x 10 x $20
= $100

P rises to $30.

C. PS on
additional units
= ½ x 5 x $10 = $25

D. Increase in PS
on initial 10 units
= 10 x $10 = $100

Producer surplus



Question

The cost of producing flat-screen TVs has fallen over the past decade. Let's consider some implications of this fact.

a. Draw a supply-and-demand diagram to show the effect of falling production costs on the price and quantity of flat-screen TVs sold.
b. In your diagram, show what happens to consumer surplus and producer surplus
c. Suppose the supply of flat-screen TVs is very elastic. Who benefits most from falling production costs − consumers or producers of these TVs?

Question

One of the largest changes in the economy over the past several decades is that technological advances have reduced the cost of making computers

a. Draw a supply-and-demand diagram to show what happened to price, quantity, consumer surplus, and producer surplus in the market for the following products.
    - Computer
    - Typewriter (Suppose that computers and typewriters are substitutes).
    - Software (Suppose that computers and software are complements)

b. Does this analysis help explain why software producer Bill Gates is one of the world's richest people?



Market Efficiency

- Total surplus = CS + PS
  - Consumer surplus = Value to buyers – Amount paid by buyers
    - Buyers' gains from participating in the market
  - Producer surplus = Amount received by sellers – Cost to sellers
    - Sellers' gains from participating in the market

Total surplus = Value to buyers – Cost to sellers

See a short scene from the movie "Pretty Woman" ("麻雀變鳳凰")



Market's Allocation of Resources

- Allocation of resources – desirable?
  - Decentralized (in a market economy)
    - Determined by interactions of many self-interested buyers and sellers
  - Total surplus – measure of society's well-being
    - To consider whether the market's allocation is efficient

If the market's allocation is generally desirable, then the role of government should be limited to the protection of property rights, national defense and so forth. If not, then public policy may potentially be able to improve upon the market's allocation.



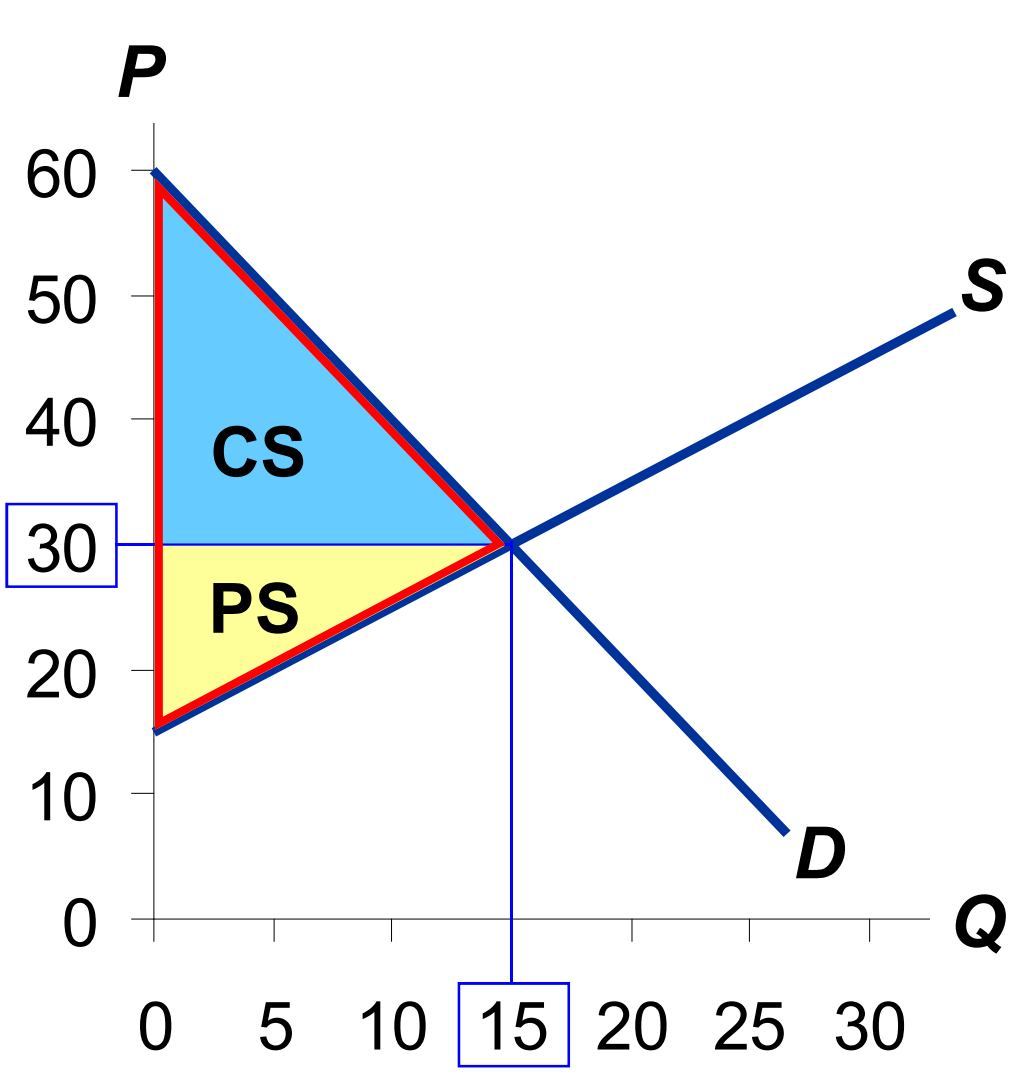
Market's Allocation of Resources

- Efficient allocation of resources maximizes total surplus
  1. The goods are consumed by the buyers who value them most highly
  2. The goods are produced by the producers with the lowest costs
  3. Raising or lowering the quantity of a good would not increase total surplus

Evaluating the Market Equilibrium

Market equilibrium:
  P = $30
  Q = 15
Total surplus
  = CS + PS

Is the market equilibrium efficient?



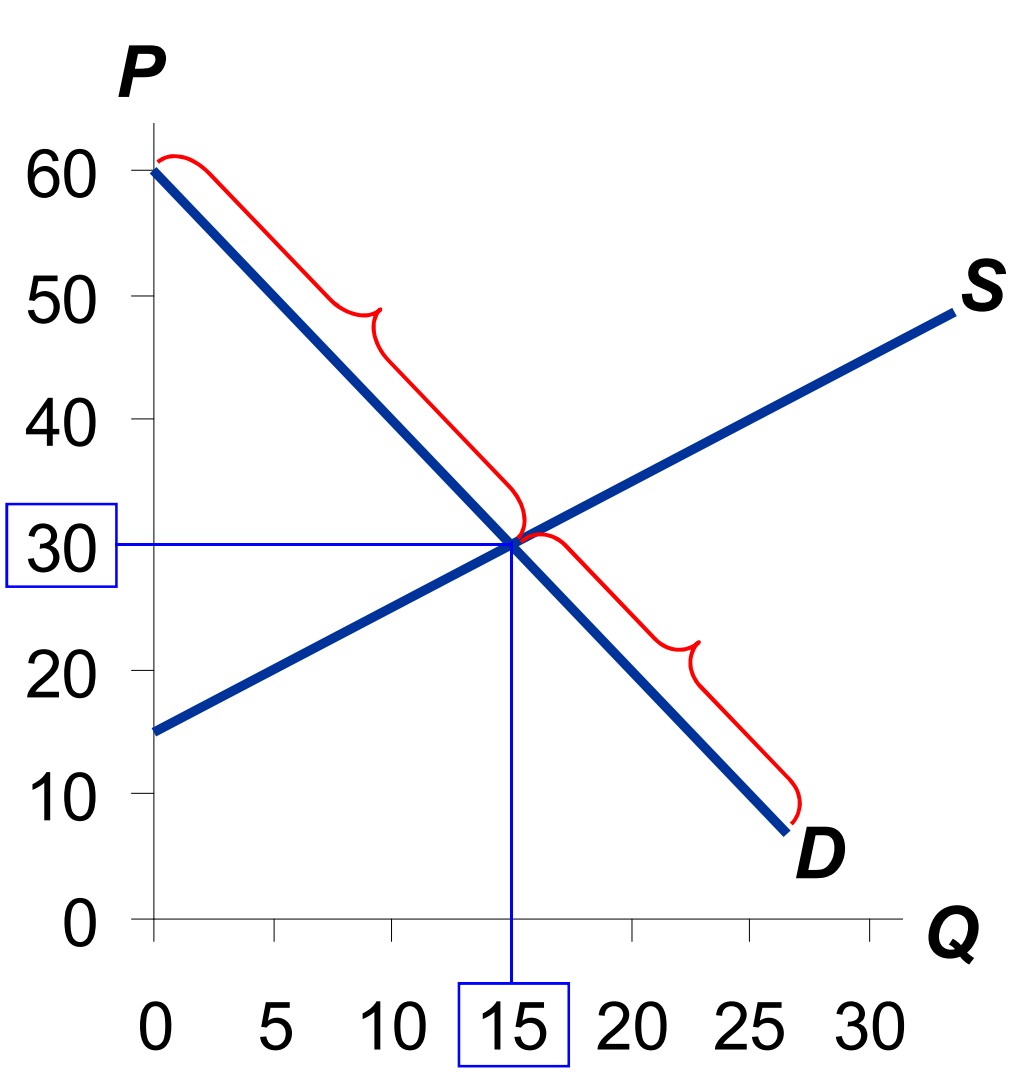
Which Buyers Consume the Good?

自由市場把商品分配給購買意願最高的那群人

Every buyer whose WTP is ≥ $30 will buy.

Every buyer whose WTP is < $30 will not.

The buyers who value the good most highly are the ones who consume it.



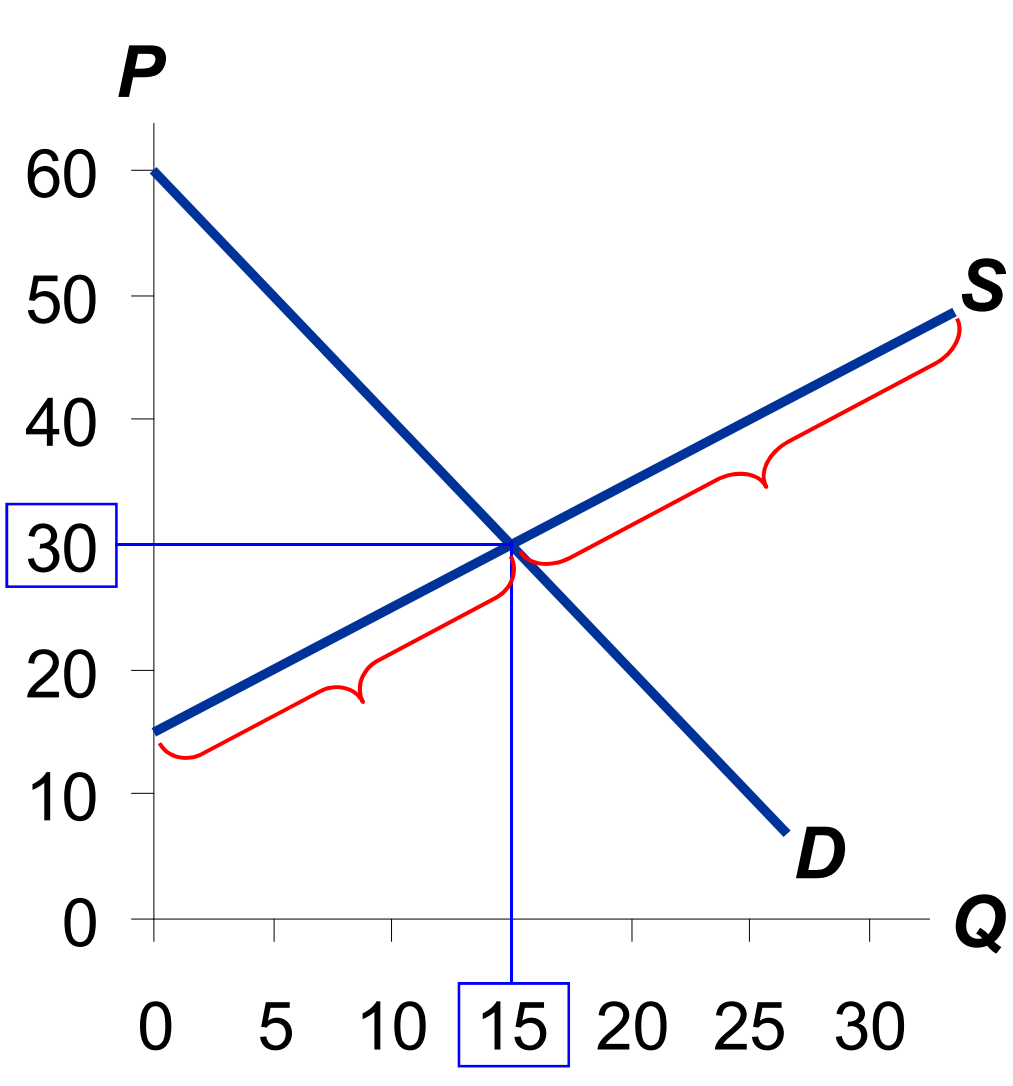
Which Sellers Produce the Good?

自由市場把商品交由生產成本最低的那群人來供應

Every seller whose cost is ≤ $30 will produce the good.

Every seller whose cost is > $30 will not.

The sellers with the lowest cost produce the good.



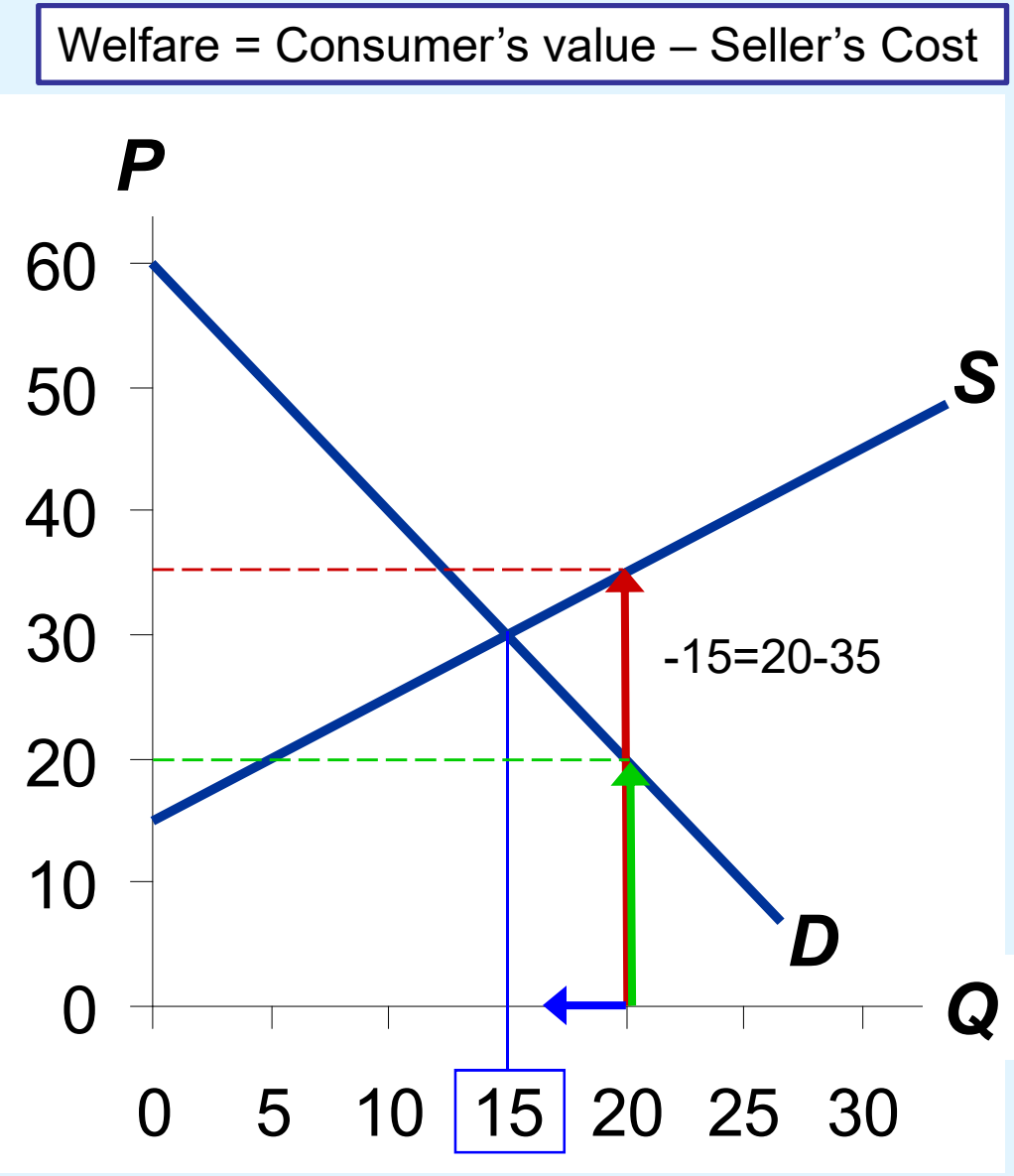
Does Equilibrium Q Maximize Total Surplus?

At **Q** = 20, cost of producing the marginal unit is $35

value to consumers of the marginal unit is only $20

Hence, can increase total surplus by reducing **Q**.

*This is true at any **Q** greater than 15.*



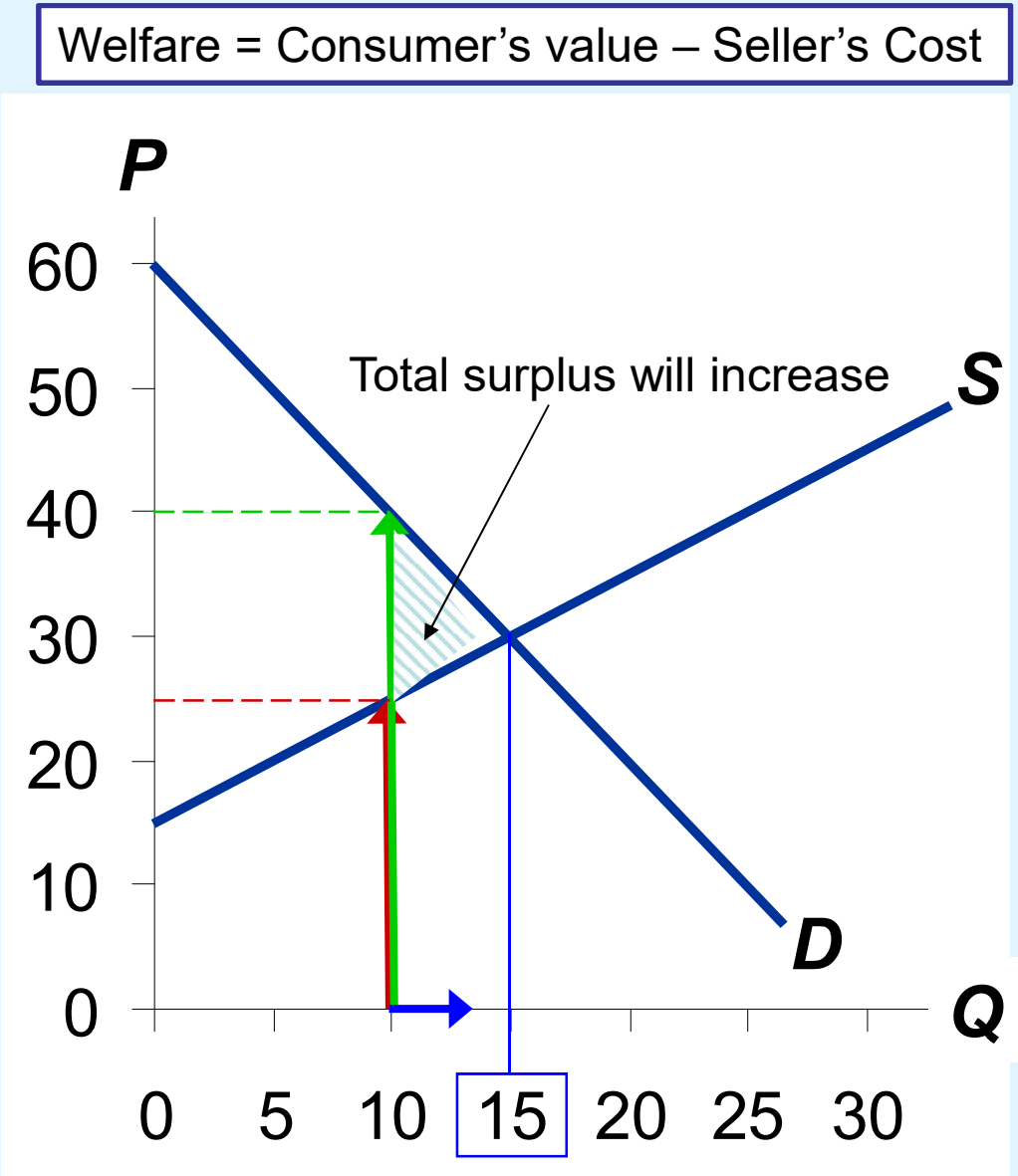
Does Equilibrium Q Maximize Total Surplus?

At ***Q*** = 10, cost of producing the marginal unit is $25

value to consumers of the marginal unit is $40

Hence, can increase total surplus by increasing ***Q***.

*This is true at any **Q** less than 15.*



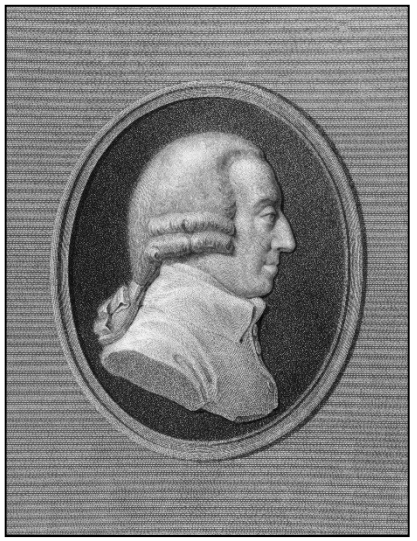


Adam Smith and the Invisible Hand
Passages from The Wealth of Nations, 1776

"Man has almost constant occasion for the help of his brethren, and it is vain for him to expect it from their benevolence only.

He will be more likely to prevail if he can interest their self-love in his favor, and show them that it is for their own advantage to do for him what he requires of them…

It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest….



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Adam Smith, 1723-1790

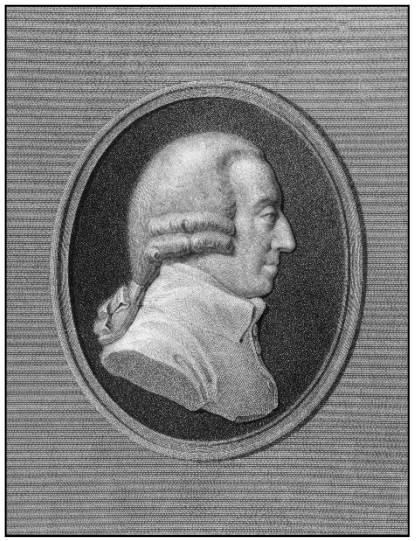


Adam Smith and the Invisible Hand
Passages from The Wealth of Nations, 1776

"Every individual…neither intends to promote the public interest, nor knows how much he is promoting it.…

He intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention.

Nor is it always the worse for the society that it was no part of it. By pursuing his own interest he frequently promotes that of the society more effectually than when he really intends to promote it."



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Adam Smith, 1723-1790



Market Efficiency

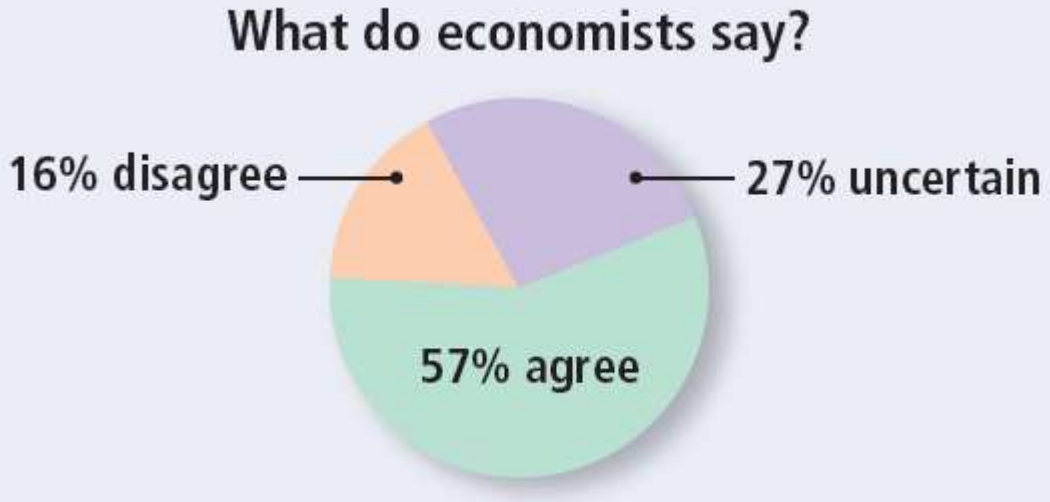
- Adam Smith's invisible hand
  - Takes all **the information** about buyers and sellers into account
  - Guides everyone in the market to the best outcome
  - Economic efficiency
- Free markets
  - Best way to organize economic activity

Economic efficiency implies an economic state in which every resource is optimally allocated to serve each individual or entity in the best way while minimizing waste and inefficiency.

ASK THE EXPERTS

Supplying Kidneys

"A market that allows payment for human kidneys should be established on a trial basis to help extend the lives of patients with kidney disease."



ASK THE EXPERTS

Supplying Kidneys

Since it is illegal to buy or sell kidneys, we are dealing with a binding price ceiling at P = $0. People that need kidney transplants get them from **friends/relatives** (no need to be on the transplant list); from **anonymous donors** (usually other people die and their family donates their organs); or stay **on the list** (sometimes for years) and getting sicker or die waiting for a kidney.

- *Q1: is the current situation efficient? (maximizes total surplus?)*
- *Q2: is the current situation fair?*
- *Q3: what would happen with the efficiency of the market if the price ceiling is removed? (e.g.: buying and selling kidneys becomes legal)*



Market Efficiency & Market Failure

- Forces of supply and demand
  - Allocate resources efficiently
- Assumptions about how markets work
  1. Markets are perfectly competitive
  2. Outcome in a market matters only to the buyers and sellers in that market
- When these assumptions do not hold
  - "Market equilibrium is efficient" may no longer be true



Market Efficiency & Market Failure

- Market failures
  - Market power: a single buyer or seller (small group) control market prices
    - Markets are inefficient
  - Externalities: decisions of buyers and sellers affect people who are not participants in the market at all
    - Inefficient equilibrium - from the standpoint of society as a whole



Should Price Gouging be Illegal?





Should Price Gouging be Illegal?





斷水斷電斷瓦斯後的民生必備物資



Eye on Price Gouging (1 of 4)

Should Price Gouging Be Illegal?

Price gouging is the practice of selling an essential item for a much higher price than normal, and usually occurs following a natural disaster.

In Florida and Texas, where hurricanes happen all too often, price gouging is illegal.

The standard economist view is that price gouging should *not* be illegal and that it is the expected and *efficient* response to a change in demand.

After a hurricane, the demand for items such as generators, and camp stoves increases and the prices of these items rise in a natural response to the change in demand.



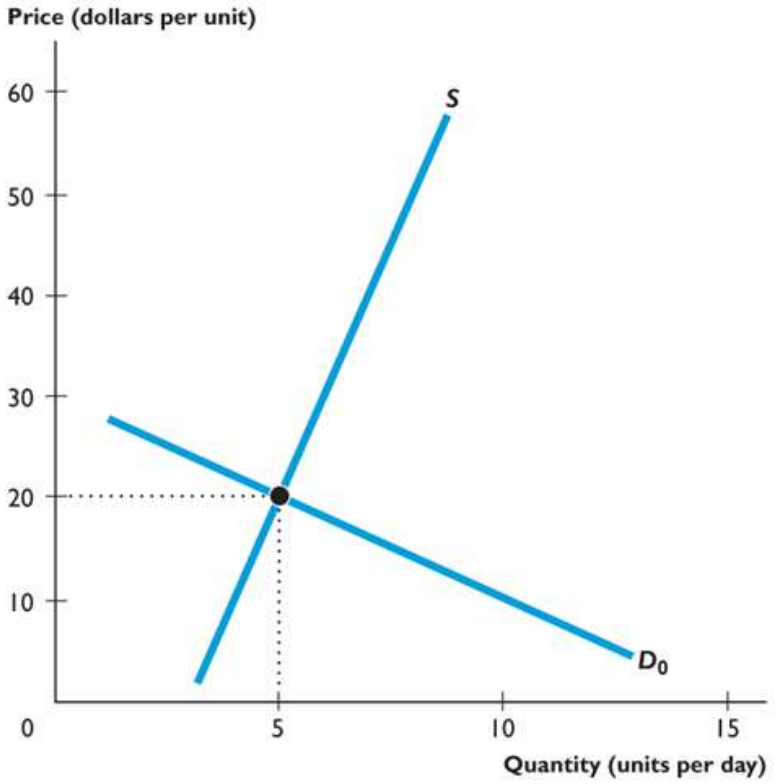
Eye on Price Gouging (2 of 4)

Should Price Gouging Be Illegal?

The figure illustrates the market for camp stoves.

The supply of stoves is the curve *S*, and in normal times, the demand for stoves is D_0

The price is $20 per stove and the equilibrium quantity is 5 stoves per day.





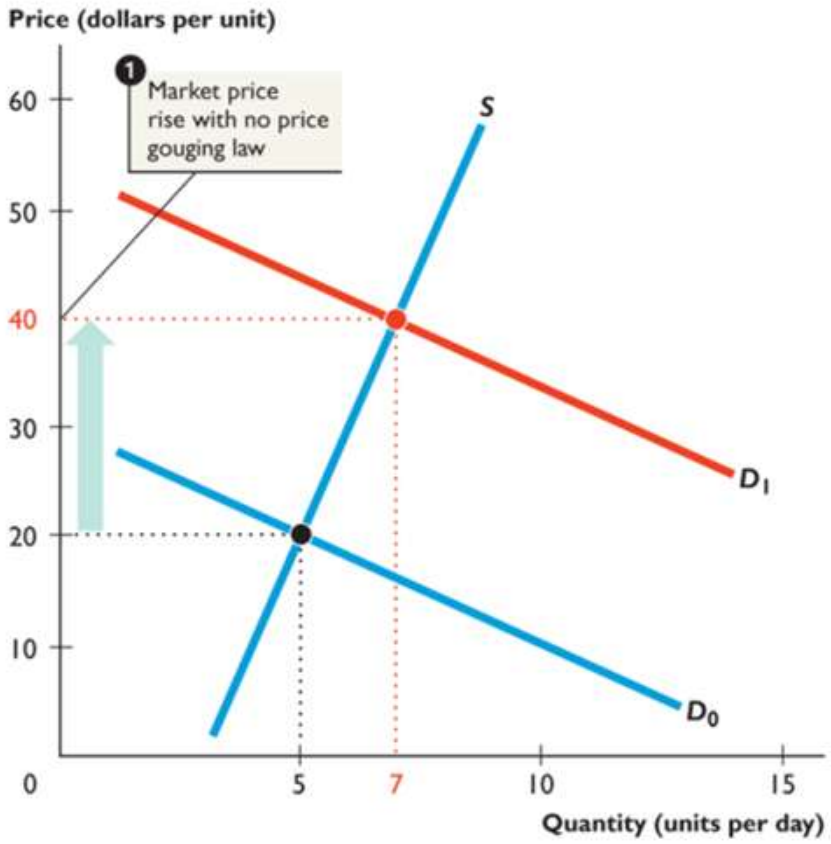
Eye on Price Gouging (3 of 4)

Should Price Gouging Be Illegal?

Following a hurricane, the demand for camp stoves increases to D_1.

With no price gouging law, the price jumps to $40 and the quantity increases to 7 stoves per day.

This outcome is efficient because the marginal cost of a stove equals the marginal benefit from a stove.





Eye on Price Gouging (4 of 4)

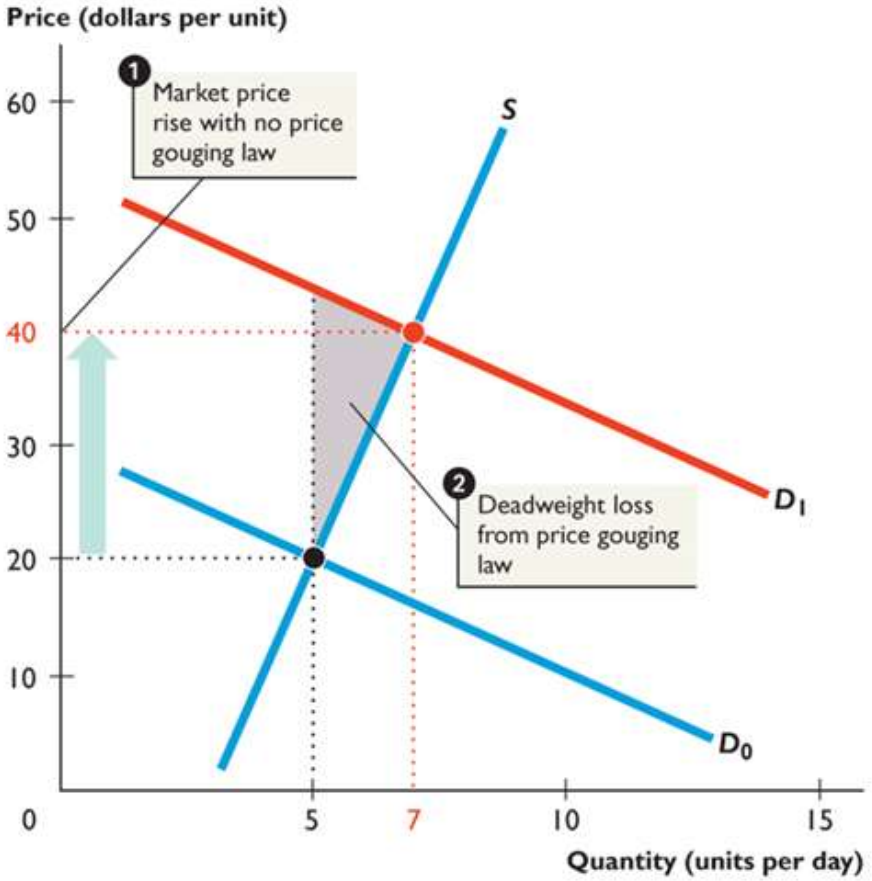
Should Price Gouging Be Illegal?

If a strict price gouging law requires the price after the hurricane to be $20 a stove, ...

Then the quantity supplied would remain at 5 stoves per day.

A deadweight loss shown by the gray triangle arises.

The price gouging law is inefficient, but is it fair?



Summary

- Consumer surplus: buyers' willingness to pay for a good minus the amount they actually pay
  - Measures the benefit buyers get from participating in a market
  - Area below the ***D*** curve and above ***P***
- Producer surplus: amount sellers receive for their goods minus their costs of production
  - Measures the benefit sellers get from participating in a market
  - Area below ***P*** and above the ***S*** curve

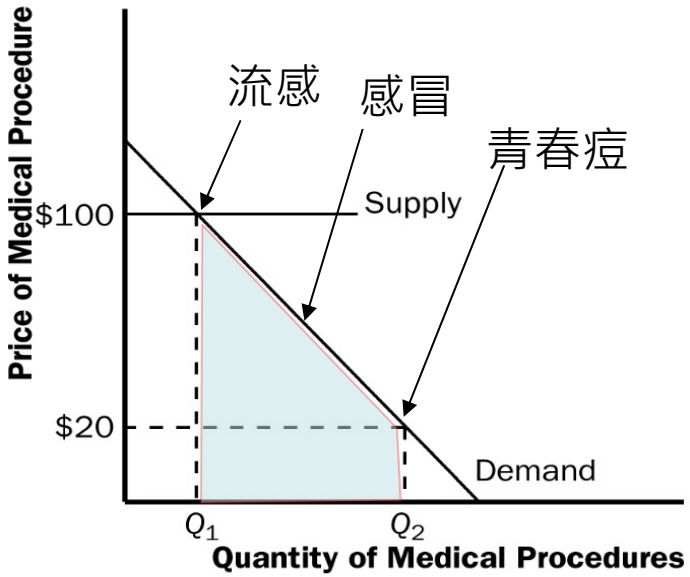
Summary

- An allocation of resources that maximizes total surplus is said to be efficient
  - Policymakers are concerned with the efficiency, as well as the equality, of economic outcomes.
- Equilibrium of ***S*** and ***D*** maximizes total surplus
  - The invisible hand of the marketplace leads buyers and sellers to allocate resources efficiently.
- Markets do not allocate resources efficiently in the presence of market failures (market power or externalities)

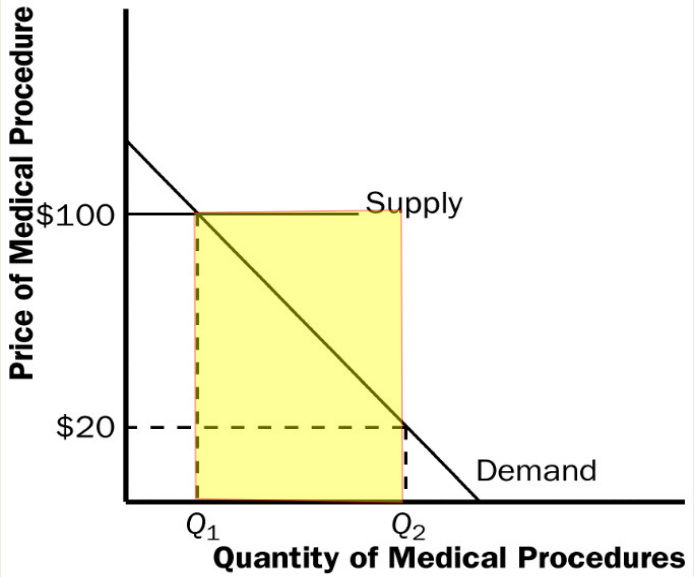
Appendix Question

- Consider how health insurance affects the quantity of healthcare services performed.
- Suppose that the typical medical procedure has a cost of $100, yet a person with health insurance pays only $20 out of pocket. Her insurance company pays the remaining $80.
  - Draw the demand curve in the market for medical care.
    - Show the quantity of procedures demanded in each procedure has a price of $100.
    - On your diagram, show the quantity of procedures demanded if consumer s pay only $20 per procedure.
  - If the cost of each procedure to society is truly $100, and if individuals have health insurance as described above, will the number of procedures performed maximize total surplus?

Reference



價值$100的保險降價到$20
消費者增加的價值(Q_1到Q_2)



價值$100的保險降價到$20
社會增加的成本(Q_1到Q_2)

Appendix Question

- Why might the use of care be viewed as "excessive"?
  - **Consumers get procedures whose value is less than the cost of producing them**. As a result, the economy's total surplus is reduced.
- What sort of policies might prevent this excessive use?
  - **The consumer must bear the marginal cost of the procedure.**
    - But this would require eliminating insurance.
  - **The insurance company could decide whether the procedure should be performed.**
    - But the insurance company does not get the benefits of the procedure, so its decisions may not reflect the value to the consumer.

Question

A friend of yours is considering two cell phone service providers.

Provider A charges $120 per month for the service regardless of the number of phone calls made.

Provider B does not have a fixed service fee but instead charges $1 per minute for calls.

Your friend's monthly demand for minutes of calling is given by an equation $Q^D = 150-50P$, where P is the price of a minute

a. With each provider, what is the cost to your friend of an extra minute on the phone?
b. In light of your answer to (a), how many minutes with each provider would your friend talk on the phone?
c. How much would she end up paying each provider every month?
d. How much consumer surplus would she obtain with each provider?
e. Which provider would you recommend that your friend choose? Why?